

The role of digital health technologies in women's health, empowerment and gender equality

Project report



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Abstract

A growing body of evidence shows that digital health technologies (DHTs) can improve women's health outcomes by overcoming ongoing socioeconomic, cultural and geopolitical barriers that inhibit their access to health care. Despite this, there has been no systematic assessment of the benefits of digital health for women's health. This project was devised to review the effects of DHTs on women's health and their potential to improve women's empowerment and gender equality by scoping peer-reviewed primary studies in this area. The results suggest that DHTs positively impact women's empowerment, facilitate the achievement of gender equality (particularly associated with improving women's access to health-care services, enhancing maternal health, providing women with essential health information and creating opportunities for women to participate in household decisions), and improve health outcomes for women (especially in relation to gynaecology, obstetrics and psychiatry), thereby advancing the achievement of the United Nations Sustainable Development Goals.

Keywords

WOMEN'S HEALTH
EMPOWERMENT
GENDER EQUITY
SUSTAINABLE DEVELOPMENT
EVIDENCE-BASED MEDICINE
QUALITATIVE RESEARCH

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Abbreviations

APA	American Psychological Association
CI	confidence interval
CINAHL	Cumulative Index to Nursing and Allied Health Literature
COVID-19	coronavirus disease 2019
DHTs	digital health technologies
Embase	Excerpta Medica Database
IT	information technology
NGO	nongovernmental organization
PRISMA-ScR	Preferred Reporting Items for Systematic Reviews and Meta-analysis Extension for Scoping Reviews
PROSPERO	International Prospective Register of Systematic Reviews
RCT	randomized clinical trial
RFO	relative frequency of occurrence
SDGs	(United Nations) Sustainable Development Goals

Executive summary

Digital health technologies (DHTs) have the potential to improve women's health outcomes for various diseases and conditions by overcoming persistent socioeconomic and geographical barriers that inhibit access to health care.

A scoping review of peer-reviewed primary studies exploring digital health and its impact on women's health and women's equality and empowerment was performed from database inception until 18 August 2023. Four databases were used – Excerpta Medica Database (Embase), MEDLINE, the American Psychological Association PsycInfo® index, the Cumulative Index to Nursing and Allied Health Literature and the Cochrane Library. Grey literature was also searched for eligible studies. Title, abstract and full-text screening, as well as data extraction, was performed by at least two independent investigators.

Forty-seven original studies evaluating and clearly reporting the effects of DHTs on women's health-related outcomes and their association with improving women's empowerment and gender equality were included. Thematic appraisal was utilized to combine qualitative data into meaningful themes and subthemes. These were mathematically converted based on the relative frequency of occurrence (RFO) of established domains, as quantifying the qualitative data through a meta-analysis of proportions enhances clarity, comparability, objectivity and the ability to draw more robust conclusions from the thematic analysis.

Overall, 37 records (RFO of 78.7% (95% confidence interval (CI): 64.3–89.3)) emphasized how the use of DHTs positively impacted women's access to health services, promoted increases in self-care and provided tailored self-monitoring systems to enable the acquisition of health-related interventions.

The applicability of DHTs to women's health, empowerment and equality achievement was observed across different medical specialties, including gynaecology and obstetrics, endocrinology and psychiatry. The utilization of digital technologies directly impacted women's empowerment and gender equality either by serving as a skills-acquisition system or a health education portal/health literacy platform (RFO of 93.6% (95% CI: 82.5–98.7); $n = 44$; 93.6%). This enabled social interaction through DHTs' potential to connect women with their environments (health-care providers and family members) (RFO of 10.6% (95% CI: 3.5–23.1)) or by improving financial-related features (by enabling cost-effective consultations and treatments or increasing financial literacy) (RFO of 10.6% (95% CI: 3.5–23.1)).

The study also highlighted the relevance of DHTs in overcoming the digital divide faced by women on a global scale within the realm of medical access and access to specialized treatment through digital interventions.

Digital modalities can overcome barriers to health care that are often associated with economic, geographic or social isolation and relative gender inequalities, allowing access to significantly vulnerable or marginalized populations.

The results presented in this review suggest that DHTs positively impact women's empowerment and facilitate the achievement of gender equality (particularly associated with improving women's access to health-care services, enhancing maternal health and providing women with essential health information, and creating opportunities for women to participate in household decisions). They also improve health outcomes among women, consequently advancing the achievement of the United Nations Sustainable Development Goals.





Introduction

Digital health technologies (DHTs) offer an enhanced mode of health-care delivery through increased access to care and a better overall experience for end-users (1). WHO defines DHTs as a discrete functionality of digital technology that are applied to achieve health objectives, which are organized into four overarching groupings based on the targeted primary user (clients, health-care providers, health systems or resources managers, and data services) (2). The Food and Drug Administration of the United States of America conceptualizes DHTs as encompassing a wide variety of technologies applicable to health, including mobile health (mHealth), health information technology (IT), wearable devices, telehealth and telemedicine, and personalized medicine (3). DHTs are also known to promote social connectivity, facilitate interpersonal communication, and create versatile working and learning opportunities for adults (4).

These technologies are not, however, equitably accessible to all sections of the population, essentially because groups that can benefit most from DHTs are often at the highest risk of digital exclusion (5,6). Research has shown how older people, women, people living with ill health or disabilities and those from lower socioeconomic backgrounds tend to experience poorer levels of Internet access and digital skills, which inadvertently affects their ability to benefit from accessing, using and engaging with DHTs (1,7,8). This digital divide distinguishes those who have access to digital-related technologies from those who involuntarily are excluded. It has been magnified due to the coronavirus disease 2019 (COVID-19) pandemic and consequent worsening of digital access (9,10). This issue becomes even more concerning, considering how DHTs are becoming integral parts of modern health systems (11–13).

Recent publications and reports have stressed the correlation between the utilization of digital technologies for women's health in the general care system and the improvement of women's empowerment (associated with terms like self-concept, self-esteem and self-worth) and equality (ensuring equality and rights in accessing education, health-care services and economic opportunities) (14–17). Substantial research has also focused on leveraging the capabilities of digital health to address needs in women's health and maternal health care by exploring the use of artificial intelligence in diagnosing endometriosis (18), using phone applications for monitoring pregnancy (19) and using digital health as a means of treating postpartum depression (20). Evidence also suggests that DHTs enable better self-management through real-time tracking of vital signs, medication adherence and lifestyle changes, and assist women in managing stress, anxiety, depression and other mental health issues (21–24).

Notably, studies focusing on North American and Middle Eastern settings have demonstrated the remarkable benefits of these technologies in not only enhancing the general health status of women patients and health-care providers (increasing efficiency and workflow optimization, improving access to information and decision support, and acting as a wellness and mental health support tool) (25,26), but also in developing an equitable and empowered environment for women's use of technologies and technology devices (26).

Statistics nevertheless emphasize that women might not fully benefit from DHTs due to the substantial gender digital divide that isolates women from technological advancements. Based on public data from the International Telecommunication Union (the United Nations specialized agency for information and communication technologies), 259 million more men than women used the Internet in 2022 (27). Although women make up about half of the global population, they comprise a disproportionate and growing share of the global offline population (27).

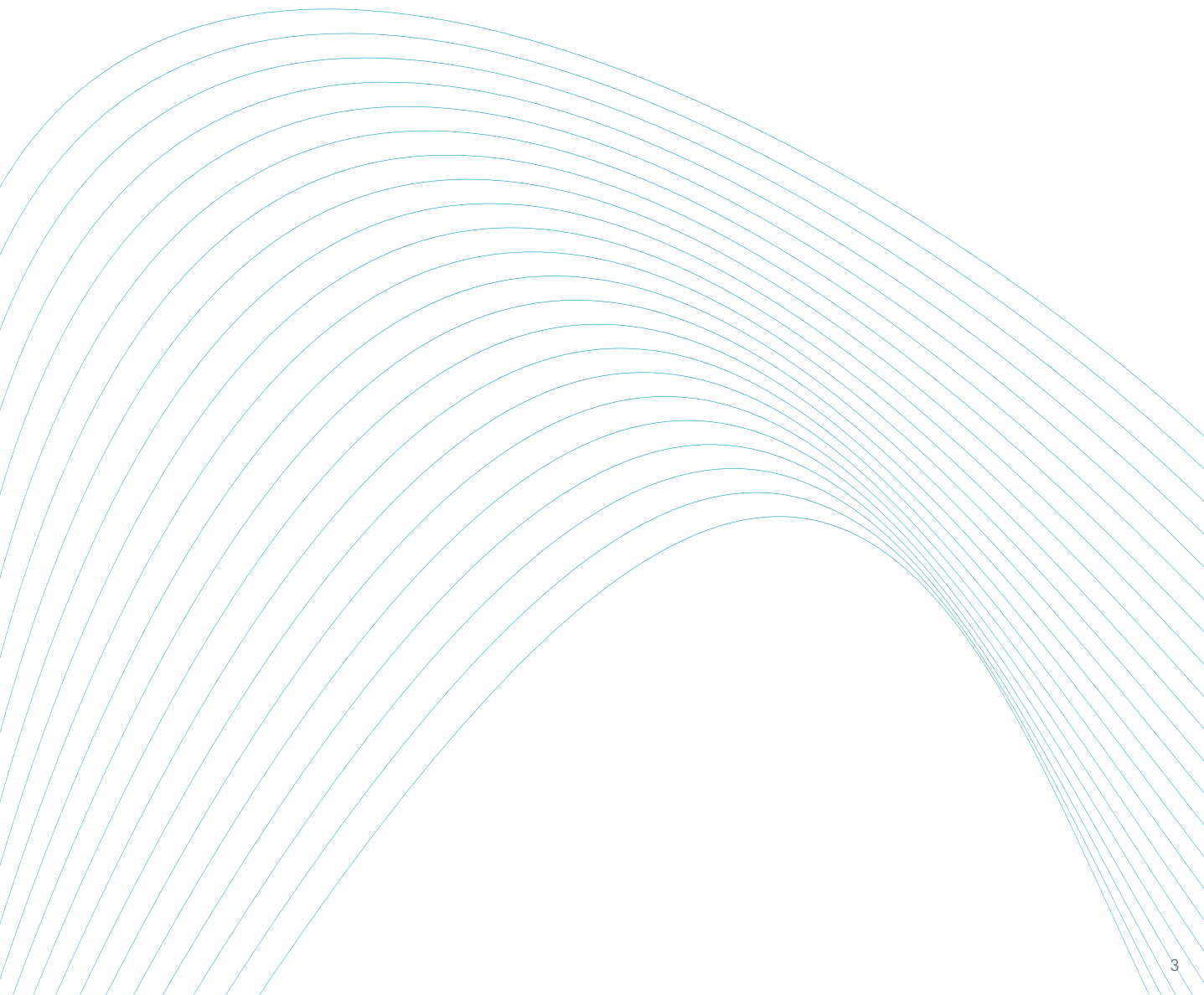
Despite studies showing the significant ability of DHTs to enhance women's health-related outcomes, challenges to implementation remain for stakeholders, health-care providers and policy-makers. Among them are accessibility and affordability (especially among marginalized or economically disadvantaged women who may lack access to smartphones, Internet connectivity or necessary devices), data privacy and security concerns, and interoperability and integration of different systems and technologies into existing health-care infrastructure (28,29). Associated challenges include the development and implementation of regulatory frameworks to oversee DHTs (30–32).

The world increasingly is embracing interconnected digital systems, but opportunities for commerce, education and health care are not uniformly expanding across all segments of society. Women often encounter numerous barriers to utilizing digital health tools, such as limited access to technologies, cultural and socioeconomic factors, and caregiving responsibilities, which exacerbate the gender digital divide and hinder the improvement of women's health and care indicators and their empowerment (33).

The United Nations established the 2030 Agenda for Sustainable Development in 2015, encompassing 17 interconnected global goals to improve health-, financial-, social- and education-related outcomes (30–32). Three of the Sustainable Development Goals (SDGs) rely on good health and population well-being (goal 3), gender equality (goal 5) and reduction of inequalities (goal 10). In consonance with the SDGs, WHO developed the *Global strategy on digital health 2020–2025* (30) and the *Regional digital health action plan for the WHO European Region 2023–2030* (34), which were designed to improve and apply DHTs in line with the vision of health for all, preserving equity, gender equality and human rights as the core elements for digital development. These global and European regional strategies set out a vision, mission, strategic objectives and framework for action to advance DHTs.

Regardless of the growing body of evidence, a systematic assessment of the benefits of digital health for women's health remains absent. No previous study, it appears, has combined evidence derived from studies assessing the impacts of DHTs on women's health status and analysing the effects of the technologies on women's empowerment and gender inequity parameters. This scoping review, covering the deployment of DHTs in the context of women's health and its potential to improve women's empowerment and gender equality in health, was therefore undertaken to provide an overview of the state of empirical research on gender equity and women's empowerment, digital health and women's health.

The main objective of the review is to provide a comprehensive synthesis of the available literature to date and determine the possible avenues through which digital health may be ready to be operationalized in practice, with the aim of facilitating a detailed plan of action for increasing women's empowerment and gender equality. The secondary objective is to summarize the direct impact of DHTs on women's health, empowerment and gender equality, and identify the factors that limit or facilitate women's access to, use of and engagement with DHTs. Factors associated with limited access and utilization of DHTs among women worldwide have also been compiled.





Methods

This review followed the methodological framework for conducting a scoping review developed by Arksey and O'Malley (35) and adheres to the Joanna Briggs Institute Manual for Evidence Synthesis for Scoping Reviews (36). The findings were reported under the Preferred Reporting Items for Systematic Reviews and Meta-analysis Extension for Scoping Reviews (PRISMA-ScR) (37). The protocol was registered on the International Prospective Register of Systematic Reviews (PROSPERO) prior to starting the review process (CRD42022350883) (38). The protocol originally was registered as a systematic review, but it was decided to pursue a broader approach and provide an overview of the key concepts within the literature at the interface of digital health and women's health. Consequently, a scoping review was performed instead of using the methodological guidelines cited above. This meant the quality of included studies was not assessed, as is common practice in scoping reviews.

Search strategy and selection criteria



Full details about the methods utilized in the synopsis are reported in the protocol, which is available elsewhere (38). Relevant scientific records were searched in four databases (the Excerpta Medica Database (Embase), MEDLINE, the American Psychological Association (APA) PsycInfo® index, the Cumulative Index to Nursing and Allied Health Literature (CINAHL) and the Cochrane Library). As suggested by Haddaway et al. (39), four searches were developed for Google Scholar based on the research questions for this review. The Google Scholar searches were conducted through Publish or Perish version 8, and the first 300 results from each search were saved and screened for eligibility.

Records from 2015 (the year the SDGs were adopted) were searched and four supplementary searches in Google Scholar were conducted, with the first 100 results from each being screened. Studies were considered eligible if they: reported the impact of DHTs and provided evidence of their effects on women's empowerment and gender equality; and defined the utilization of DHTs on health-related outcomes. Studies enrolling transwomen were deemed eligible, although it is important to note that their unique health status and needs may not have been recognized.

Publications were not excluded based on published language, but conference proceedings, editorials, correspondence, short communications, studies comparing male and female populations in their analyses (Box 1), studies that did not include health-related data, and data not published in the format of an original research study were excluded. A full definition of terms associated with the review, including the types of DHTs and the review's used terms, is presented in Annex 1.

Box 1. Excluding studies comparing women and men in their assessment

Studies enrolling both women and men in their methodological design were excluded because:

- ▶ women regularly have unique basic health needs and experiences that may not be effectively approached in studies that enrol both sexes collectively;
- ▶ currently, there are disparities in health-care access, treatment and associated outcomes among men and women – prioritizing studies considering women only permits for a more specialized and nuanced understanding of how DHTs assist the reduction of such disparities and improve women's health-related outcomes;
- ▶ DHTs tailored specifically for women might appraise issues such as reproductive health, maternal care, menstrual health or health-related conditions that are predominantly applicable to women, including breast cancer and osteoporosis – deleting records enrolling both sexes enabled interventions tailored explicitly to women's health needs to be targeted;
- ▶ understanding how DHTs empower female patients, enabling them to actively control their health, make informed decisions and access health-care services, can significantly influence gender equality – these interventions might impact women's empowerment in managing their health and well-being; and
- ▶ evaluating studies solely focused on female patients enables a more detailed and specific examination of outcomes, experiences and challenges faced by women in utilizing DHTs.

Following duplicate removal (using EndNote™ 20 (Clarivate™, St Helier, Jersey)), obtained records were uploaded into Covidence® (Veritas Health Innovation, Melbourne, Australia), primarily for the title and abstract screening phase and subsequently to the full-text screening stage. Screening processes were performed independently by eight review authors (Borges do Nascimento, Abdulazeem, Marquez, Weerasekara, Kularathne, Østeengaard, Vasanthan and Novillo-Ortiz). Any decisional disparity was resolved by group discussion.

Search-string levels associated with the digital divide included “Digital Divide”, “Digital Equality”, “Digital Inequality” and “Digital Gap”, “Gender Equity”, “Women's Rights” and their synonyms. Elements related to DHTs mostly were represented by key terms such as “Digital Technology” and “Information Technology”, “Telemedicine”, “Telehealth” and “Mobile Health”.

The searches in the databases were performed on 18 August 2023 by an experienced librarian from the Cochrane team. The search strategy (Annex 2) was designed in collaboration with a group of experts and the information specialist. Bibliographies of records shortlisted for full-text analysis were evaluated to ensure an exhaustive search.

Data extraction



Each record underwent full data extraction managed by eight review investigators. Elementary data from included studies were abstracted, including: study identification; publication year; journal name; study design; study objective; the number of included individuals; and setting and country in which the study was carried out. As outlined in the protocol, the extracted and evaluated primary outcomes consisted of: the type of DHT utilized; medical conditions and health issues being assessed; and the main findings raised in each publication. Data extracted from each included study were assessed by at least two investigators. If any data extraction conflict occurred, a discussion between the conflicting investigators was held to define which elements should be displayed in the final materials. Data management and recording were performed through an online Microsoft Excel file (Microsoft Corporation, Redmond (WA), United States of America).

Data analysis



The protocol stated that only a descriptive assessment of the obtained data would be carried out, allowing the simultaneous inclusion and analysis of quantitative and qualitative studies. A mathematical model to report findings from qualitative research based on thematic analysis concepts was developed as part of the research group's scientific methodology. The initial main findings of the included studies were coded and thematically collated using NVivo® software (QSR International, Burlington (VT), United States) by two investigators and subsequently re-evaluated by two other researchers. At least four researchers therefore participated in the combination of findings into similar descriptive themes. Geographical data were summarized and mapped using Quantum Geographic Information System software.

After establishing analytical themes, the frequency of occurrence for each categorized higher domain was aggregated into a standard meta-analysis of proportions. To better illustrate how the knowledge provided within the results section of primary studies was translated into the created domains, statements were randomly selected from these manuscripts to justify the allocation for the determined theme ([Table 1](#)).

Table 1. Primary studies, with the rationale utilized for thematic categorization of findings

Author/year	Country	Quotation	Theme addressed
Slomian et al., 2017 (40)	Belgium	"The majority of the women considered the information that they found on the Internet to be 'somewhat useful' (72.5%) or 'useful' (10.2%). When asked about finding the information about themselves on the Internet, most women reported they 'most of the time' (74.2%) or 'always' (9.10%) found the information they needed; 16.3% reported they 'rarely' found the information they needed, and only one woman reported 'never' finding information on the topic she was searching for. When asked about finding information about their child on the Internet, most women reported they 'most of the time' (78.0%) or 'always' (6.90%) found the information they needed; 13.8% reported they 'rarely' found the information they needed, and three women reported 'never' finding information on the topic they were searching for."	The content of the passage was associated with the "use of content to improve health-related outcomes, serving as a skills-acquisition system, health education and literacy portal" theme.
Slomian et al., 2017(40)	Belgium	"Searching information on the Internet seemed to help many women in their decision-making process. Approximately 80% of the women felt that the Internet helped them control a decision that they made 'little', 'often' or 'very often'. The women were asked to rate their confidence in decision making (scale of 0–10) for their own health and for their child's health both before and after using the Internet. There was a significant increase in the median confidence for those two scales: from 6.7 (4.9–8.1) before to 7.1 (5.5–8.0) after using the Internet for decisions about their own health ($p = 0.003$) and from 6.2 (4.5–7.5) before to 6.7 (5.1–7.8) after using the Internet for decisions about their child's health ($p < 0.0001$)."	The content of the passage was associated with the "increase women's autonomy and enhance decision-making capacity" theme.
Dalal et al., 2022 (41)	India	"Information and Communication Technologies or ITC has an important role to play in the achievement of gender equity as well as women's empowerment, although it might amplify the existing inequity, because of lower usage of digital technologies by women ... only electronic empowerment, without the provision of education cannot bring about a considerable change in the prevalence of intimate partner violence ... against women in India."	The content of the passage was associated with the "use of content to retrieve women's rights and against violence, including the decision of their best idea of body image" theme.

Table 1 contd

Author/year	Country	Quotation	Theme addressed
Chowdhury et al., 2019 (42)	Bangladesh	"Our study also demonstrated that Aponjon [an evaluated mobile health service] can increase knowledge on the symptoms of pregnancy-related complications. We found greater knowledge about signs of eclampsia and initiatives to be taken for different birth preparedness components among Aponjon service-users compared to non-users."	The content of the passage was associated with the "use of content to improve health-related features, serving as a skills-acquisition system, health education and literacy portal" theme.

A framework developed by Morgan et al. (43) was used for data summarization of gender equality and women's empowerment outcomes concepts. Women's empowerment aims to address gender inequality (or unequal power relations) by enabling women to gain more control over their lives and to shape systems around them, such as the organization and delivery of health systems. This gender framework explores the ways in which gender power relations manifest as inequities through inequitable access to resources, roles and practices, norms and beliefs, and decision-making power and autonomy. Based on the identified barriers for utilization of, and access to, DHTs by women, potential actions for multiple stakeholders to optimize the effects of such technologies in practice were outlined. These potential actions are presented in the discussion chapter.

Nominally identified results are indicated as the relative frequency of occurrence (RFO) and 95% confidence interval (CI). Analysis was executed using R software version 4.1.1 (R Foundation for Statistical Computing, Vienna, Austria), using the metapro function package (44). This graphic and mathematical presentation of findings was created for several reasons. First, it is unquestionable that qualitative data are powerful in presenting the details of research, but the generated results might be difficult to aggregate and interpret. By using the adopted methodology, categorized qualitative data could be converted into numerical results, allowing a more straightforward and quantitative evaluation of the findings.

Secondly, mathematical models (such as the proposed proportion-based meta-analysis) provide an evaluation of qualitative data. By enabling a standardized analysis and presentation of data, the inherent bias with subjectivity is reduced. Lastly, the strong assumption was that quantitative representations of qualitative data can be more accessible and easily understandable to the average reader (either from the medical or technology industries). It enables clearer communication of complex qualitative findings to average readers who might be more used to quantitative results.

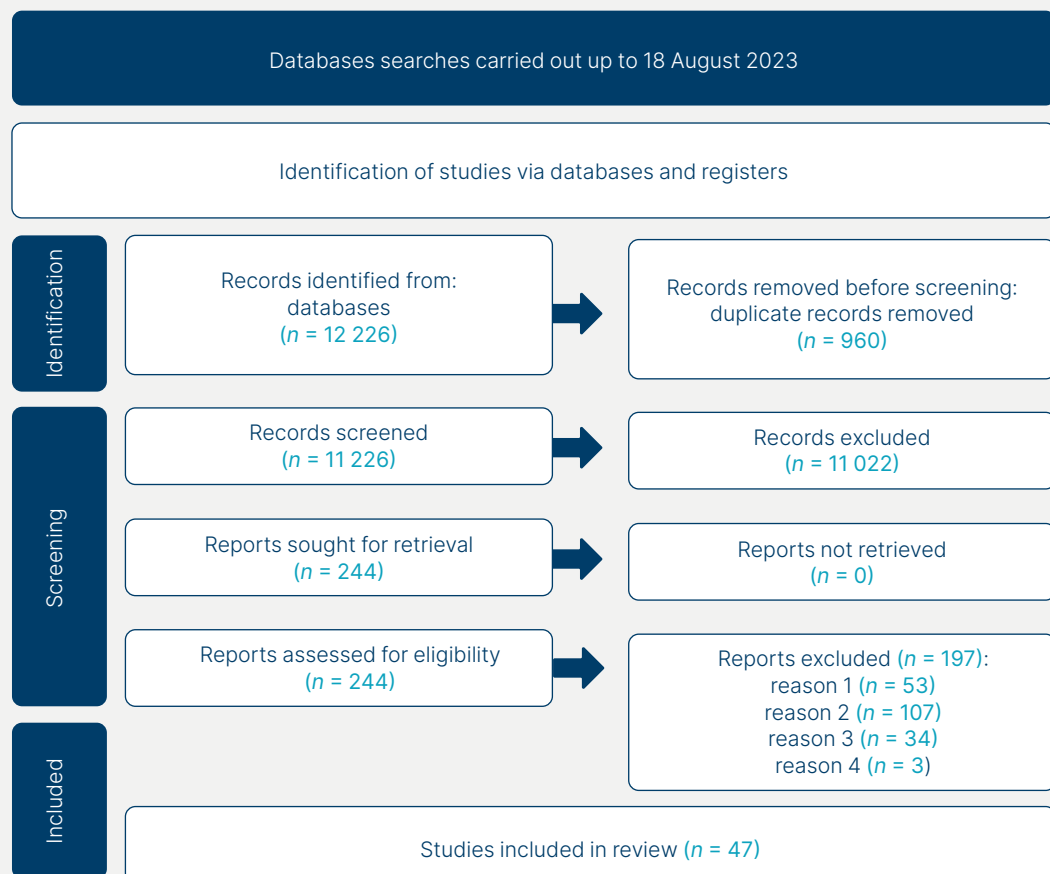
This study was deemed exempt from ethical approval as it did not assess data or intervene with humans.



Results

The search performed by the panel of specialists identified 15 404 records for screening. After duplicate removal, 9452 studies were evaluated during the title and abstract assessment. Ultimately, 47 original studies evaluating and clearly reporting DHTs on women's health-related outcomes and their association with improving women's empowerment and gender equality were included (40–42,44–88). Fig. 1 shows the review's flow diagram. A complete list of excluded studies with the justification for exclusion is presented in Annex 3.

Fig. 1. Review's flow diagram



Reason 1: wrong study design or compared women with men.

Reason 2: studies not providing any specific and remarked data on how DHTs could improve gender equity and women's empowerment outcomes.

Reason 3: conference abstracts.

Reason 4: no digital intervention was highlighted in the manuscript.

Most of the included primary studies were surveys, in-depth interviews, focus-group studies and observational studies (40–42,45–52,54–59,61–63,66,68–73,75,76,78–80,83–88). Nine (19.1%) were designed as randomized clinical trials (RCTs) (53,60,64,65,67,74,77,81,82). The study design and main characteristics of the included studies are summarized in Table 2. As far as the countries and regions represented are concerned, included studies represented all main global continents, although disproportionately. Table 3 shows the number of studies indicating the DHT implications per WHO region and countries/regions. The most represented countries in terms of occurrence were the United States and the United Kingdom ($n = 6$; 12.7%).

Table 2. Main characteristics of included studies

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Abrejo et al., 2022 (48)	<i>BMC Women's Health</i>	Pakistan	To explore the feasibility and acceptability of potential mHealth intervention for women living in low socioeconomic areas to increase uptake of family planning	30 (23 married women and seven health-care professionals)	In-depth interviews
Aiken et al., 2017 (49)	<i>British Journal of Obstetrics and Gynaecology</i>	Ireland and United Kingdom (Northern Ireland)	To examine the characteristics and experiences of women seeking at-home medical termination of pregnancy using online telemedicine	1 023	Online questionnaire
Alabdullatif et al., 2022 (50)	<i>International Journal of Environmental Research and Public Health</i>	United States	To assess the association between IT-based health-care communication and mammography utilization and to evaluate if this effect is modified by race/ethnicity and age	94 290	Survey
Ampt et al., 2020 (51)	<i>JMIR mHealth and UHealth</i>	Kenya	To describe the process of development of the WHISPER intervention, its theoretical framework, key domains and strategies, and to explore participants' responses to the proposed intervention regarding message content, behaviour change construct, feasibility and acceptability	42	In-depth interviews

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Atukunda et al., 2021 (52)	<i>JMIR Formative Research</i>	Uganda	To develop a novel patient-centred mHealth intervention to encourage and support women to use maternity care services	35 (30 women and five health-care providers)	In-depth interviews
Atukunda et al., 2023 (53)	<i>JMIR Formative Research</i>	Uganda	To evaluate the feasibility, acceptability and preliminary efficiency of a novel mHealth-based messaging app to promote maternity services among pregnant women	120	RCT
Boydell et al., 2021 (54)	<i>British Journal of Obstetrics and Gynaecology</i>	United Kingdom (Scotland)	To explore the experiences of women who accessed medical abortion at home up to 12 weeks gestation, delivered via telemedicine abortion service implemented in response to the COVID-19 pandemic to identify areas for improvement and inform service provision	20	In-depth interviews
Camacho-Miñano et al., 2021 (55)	<i>Journal of Gender Studies</i>	Spain	To explore how pedagogies of perfection in fitness content on social media work as a postfeminist technology of the self, exhorting young women to act upon themselves to become perfect but hiding the extent to which exercise as aesthetic labour is normatively demanded	37	Focus-group and in-depth interviews
Choo et al., 2015 (56)	<i>Addiction Disorders & their Treatment</i>	United States	To explore women's attitudes to use of computers for screening and intervening in drug use and partner abuse	17	In-depth interviews

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Chow et al., 2020 (57)	<i>JMIR Cancer</i>	United States	To evaluate the acceptability and preliminary efficacy of a novel app-based intervention with phone coaching in survivors of women's cancer	28	Focus-group analyses
Chowdhury et al., 2019 (42)	<i>BMC Medical Informatics and Decision Making</i>	Bangladesh	To evaluate the effects of Aponjon use on knowledge and behaviours related to maternal and newborn health care	6 551	Retrospective observational study
Daehn et al., 2023 (58)	<i>BMC Pregnancy and Childbirth</i>	Germany	To evaluate the feasibility of SmartMoms (German-based web app) to inform women about postpartum depression, support them and provide an easily accessible self-screening instrument	217	Mixed-methods approach (online surveys, usage data and interviews)
Dalal et al., 2022 (41)	<i>BMC Women's Health</i>	India	To investigate whether economic and electronic empowerment of women act as protective factors against intimate-partner violence	66 013	Survey
Da Silva et al., 2019 (59)	<i>Revista Brasileira de Enfermagem</i>	Brazil	To evaluate the GestAcao application, based on the experience of pregnant women in prenatal consultations in public health services	13	Prospective observational study
Derksen et al., 2023 (60)	<i>Applied Psychology: Health and Well-being</i>	Germany	To test a short digital communication intervention based on the health action process approach	360	RCT

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Doherty et al., 2018 (61)	<i>JMIR Mental Health</i>	United Kingdom	To explore the issues and challenges surrounding the use of mobile phones for self-support of psychological well-being during pregnancy	38	Qualitative interviews
Ellis & Roberts, 2020 (62)	<i>Birth: Issues in Perinatal Care</i>	United Kingdom	To explore online pregnancy forum health-related use and evaluate quality of information shared	480 discussion threads	Retrospective observational study
Garcia et al., 2019 (45)	<i>Journal of Medical Systems</i>	Spain	An explorative study to improve the mental, physical and social health of vulnerable immigrant women	71	Exploratory study
Gilmour et al., 2017 (63)	<i>Health Informatics Journal</i>	Canada	To test the feasibility of self-care web applications to inform women if, when and where to seek help for symptoms after hysterectomy	31	Survey and in-depth interviews
Gitonga et al., 2021 (64)	<i>African Journal of Midwifery and Women's Health</i>	Kenya	To examine the influence of short message-service reminders on utilization of focused antenatal care	118	RCT
Hackett et al., 2018 (65)	<i>PLoS ONE</i>	United Republic of Tanzania	To evaluate a smartphone-based application designed to assist community health workers with data collection, education delivery, gestational danger sign identification and referrals	572	Cluster RCT

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Heitmann & Schjøtt, 2020 (66)	<i>Maternal and Child Health Journal</i>	Norway	To describe the development and future perspectives of an innovative medicines-information service aiming to increase empowerment among pregnant and breastfeeding women	30 000 answered questions	Descriptive study
Jones et al., 2021 (67)	<i>Drug and Alcohol Dependence</i>	United States	To evaluate the feasibility and acceptability of two sex and female empowerment interventions (face-to-face and computer-adapted) compared to usual care and their efficiency to improve contraception utilization	70	Pilot RCT
Killinger et al., 2022 (68)	<i>BMJ Sex and Reproductive Health</i>	Germany	To understand the motivations and perceived barriers to access for women who choose telemedicine abortion outside formal sector	1 090	Mixed-method study
Kodama et al., 2022 (69)	<i>Journal of Psychosocial Nursing and Mental Health services</i>	Japan	To investigate pregnant women's use of pregnancy apps and how these affect their mental health	110	Survey
Laar et al., 2019 (70)	<i>Public Health</i>	Ghana	To assess the perspectives of women and health workers on the feasibility of mobile health technology for the provision of maternal and child health services in rural settings	489	Mixed-method study

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Lackie et al., 2021 (71)	<i>Journal of Medical Internet Research</i>	Canada	To engage women from a variety of backgrounds and locations to determine the unmet psychoeducational needs of women with postpartum depression and how a web-enabled platform used to deliver psychosocial skills and education to assist in the management could fulfil those needs	31	Focus group
Lovell & Harris, 2021 (72)	<i>Midwifery</i>	United Kingdom	To examine women's use of mobile apps in labour	749	Survey
Lupton & Maslen, 2019 (46)	<i>Journal of Medical Internet Research</i>	Australia	To investigate which types of digital technologies women used regularly for health-related purposes and which they found most helpful and useful	66	Focus-group and in-depth interviews
Mackintosh et al., 2020 (73)	<i>Midwifery</i>	United Kingdom	To assess the role of online resources and apps for women's help-seeking and staff responses to concerns in the perinatal period	1 254	Survey
Malachi et al., 2023 (74)	<i>African Journal of Nursing and Midwifery</i>	Kenya	To evaluate the effectiveness of health education on antenatal care provided through mobile phone calls among pregnant women	160	RCT

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
McBride et al., 2018 (75)	<i>Journal of Public Health</i>	Viet Nam	To pilot and evaluate a low-cost mHealth intervention called “mMom” utilizing behaviour change communication to improve access to maternal, newborn and child health services and health equity among ethnic minority women living in remote areas	12	Prospective observational study
Musiimenta et al., 2020 (76)	<i>Online Journal of Public Health Informatics</i>	Uganda	To explore how illiterate pregnant women perceive a maternal health mobile application composed of tailored video and audio messages, appointment reminders and calling function	14	Mixed-method study
Ngo et al., 2022 (77)	<i>JMIR Mhealth Uhealth</i>	Norway	To investigate whether the MinSafeStart mobile app could impact nausea- and vomiting- related symptoms, quality of life and decisional conflict regarding nausea and vomiting treatment	268	RCT
Özkan Şat & Yaman Sözbir, 2021 (78)	<i>Maternal and Child Health Journal</i>	Türkiye	To identify the use of mobile applications by pregnant women in receiving health information, counselling and health care during the COVID-19 pandemic and their distress levels during pregnancy	376	Survey

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Perrenoud et al., 2022 (47)	<i>Women and Birth</i>	Switzerland	To analyze how health and social care professionals (with a focus on community midwives and women) communicate between postpartum home visits through immediate message application	50 (30 health-care professionals and 20 immigrant women)	Socio-anthropological study
Rajkhowa & Qaim, 2022 (79)	<i>Social Science & Medicine</i>	India	To analyse associations between women's use of mobile phones and selected indices of female autonomy and empowerment	39 523	Survey
Riley & Paskova, 2022 (80)	<i>Digital Health</i>	New Zealand	To propose a theoretical framework for understanding women's use of menstrual-tracking digital applications and apply it to the analysis of users' experiences in the management of premenstrual syndrome	5	In-depth interviews
Roh et al., 2023 (81)	<i>JMIR Formative Research</i>	United States	To assess the feasibility and efficacy of the "wMammogram" intervention, to undergo breast cancer screening as compared with the control group (educational brochure)	122	RCT
Rollo et al., 2020 (82)	<i>International Journal of Environmental Research and Public Health</i>	Australia	To evaluate the feasibility, including recruitment, retention, preliminary efficacy and acceptability of the "Body beyond balance" eHealth intervention in women with previous gestational diabetes mellitus	42	RCT

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Slomian et al., 2017 (40)	<i>Midwifery</i>	Belgium	To evaluate the need for information after childbirth and what questions do new mothers ask themselves; to assess why and how women use the Internet to meet their needs for information; to describe how respondents evaluate the reliability of the information found; to understand how the information found on the Internet affects women's decision-making; and to appreciate how health professionals react to the information found by the women	349 questionnaire answers	Survey
Singleton et al., 2022 (83)	<i>Journal of Cancer Survivorship</i>	Australia	To evaluate the reach, usefulness, acceptability and factors influencing engagement with a lifestyle-focused text-message intervention to support women's mental and physical health after breast cancer treatment	160	Mixed-method study
Thompson et al., 2020 (84)	<i>mHealth</i>	United States	To examine the feasibility of a programme that aimed to help women with opioid-use disorder to meet their reproductive health needs by bringing services directly to them by using community outreach educators and telehealth	51	Prospective observational study

Table 2 contd

Study ID	Journal	Country/ region	Objective(s)	Sample size	Study design
Ure et al., 2020 (87)	<i>Journal of Medical Internet Research</i>	United Kingdom	To explore how women use and experience social media to self-manage their psychosocial needs and support self-management across the breast cancer continuum	21	In-depth interviews
Wagg et al., 2019 (85)	<i>Heliyon</i>	United Kingdom	To document and describe the posts made within an online breastfeeding support group	501 posts	Qualitative study
Webber et al., 2020 (86)	<i>Women's Health</i>	United Republic of Tanzania	To improve maternal health and women's access to maternity care services	112 (31 women and 43 health-care professionals)	Focus group
Zhang et al., 2022 (88)	<i>JMIR Formative Research</i>	China	To investigate the use of maternal and child health apps among pregnant women and explore associations with pregnancy outcomes	1 850	Retrospective observational study

Table 3. Number of studies indicating the DHT implications by WHO region and countries/regions

WHO region	Country/region	Number						
		A	B	C	D	E	F	G
Africa	Bangladesh	1	–	–	–	–	–	–
	Ghana	–	–	–	–	1	–	–
	Kenya	3	–	–	1	1	–	1
	Uganda	3	–	1	–	3	–	3
	United Republic of Tanzania	2	–	–	–	–	–	–
Americas	Brazil	1	–	–	–	–	–	1
	Canada	2	–	–	–	2	–	–
	United States	5	–	–	1	2	1	3
Eastern Mediterranean	Pakistan	1	–	–	–	1	1	1
Europe	Belgium	1	–	–	–	1	1	–
	Germany	2	–	1	–	1	2	3
	Ireland and United Kingdom (Northern Ireland)	1	–	1	1	–	–	–
	Norway	2	–	–	–	1	–	1
	Spain	2	1	–	1	1	–	1
	Switzerland	1	1	–	–	1	–	1
	Türkiye	1	–	–	–	–	–	–
	United Kingdom	6	–	–	–	3	1	1
	United Kingdom (Scotland)	1	–	1	–	–	1	1
South-East Asia	India	2	1	1	1	2	–	1
Western Pacific	Australia	3	2	–	2	2	1	2
	China	1	–	–	–	–	–	–
	Japan	1	–	–	–	–	–	–
	New Zealand	1	–	–	–	–	1	1
	Viet Nam	1	–	–	–	1	–	1

A: use of content to improve health-related features, serving as a skills-acquisition system, health education and literacy portal.

B: use of content to improve social-related features, serving as a connection with the environment and allowing participation.

C: use of content to improve financial-related features.

D: use of content to retrieve women's rights and against violence, including the decision on their best idea of body image.

E: improved household relationships by allowing family-planning discussions or stating their needs in health-care settings, serving as a communication mechanism.

F: enhanced the feeling of speaking out about daily problems, allowing sharing of feelings without stigmatization or fear.

G: technology was considered to increase women's autonomy and enhance their decision-making capacity, either by the access facility, flexibility, security and confidentiality of usage, or by offering a sense of ownership and inclusiveness.

The included studies predominantly were associated with mobile health (including the use of applications and short-message services: $n = 30$; 63.8%) (41,42,45,47,48,51–53,57,59–61, 64–66,69,70,72–81,83,86,88). Telehealth (including telemedicine) and web-based platforms were described in 13 primary studies (27.6%) (40,41,46,49,50,54,56,58,63,67,68,71,84). Social media platforms (including online discussion forums and groups: $n = 5$; 10.6%) (41,55,62,85,87) and digital health systems ($n = 1$; 2.1%) (82) were identified among the included records.

Overall impact of DHTs on women's health status



As stated in the methods chapter, included studies reported data on how, and to what extent, DHTs improved women's health status. Table 4 summarizes the relative frequency of categorized domains identified among included studies regarding the impact of DHTs on women's health.

Table 4. Summary of findings (health-related outcomes)

Outcomes (effect of DHTs)	RFO % (95% CI)	Number of participants [studies]	Comments
Improved psychiatric outcomes	44.7 (30.2–59.9)	72 759 [21]	In addition, 480 discussion threads and 501 posts were considered as the sample.
Induced psychological discomfort or negative emotional burden	10.6 (3.5–23.1)	1 881 [5]	–
Improved gynaecological and obstetric outcomes	53.2 (38.1–67.9)	144 841 [25]	In addition, 349 answered questionnaires were considered as the sample.
Increased health access, self-care and self-monitoring	78.7 (64.3–89.3)	54 570 [37]	In addition, 480 discussion threads, 349 answered questionnaires and 501 posts were considered as the sample.
Enhanced metabolic or endocrinal outcomes	10.6 (3.5–23.1)	2 405 [5]	–
Helpful in disease prevention (including infectious diseases) and health promotion	25.5 (13.9–40.3)	162 954 [12]	–
Useful for appropriate use of medications	19.1 (9.1–33.3)	95 426 [9]	In addition, 30 000 answered questions and 501 posts were considered as the sample.

Table 4 contd

Outcomes (effect of DHTs)	RFO % (95% CI)	Number of participants [studies]	Comments
Helpful in sexual and gynaecological health	53.2 (38.1–67.9)	144 585 [25]	In addition, 349 answered questionnaires were considered as the sample.
Increased identification of medical-related issues	19.1 (9.1–33.3)	8 829 [9]	In addition, 349 questionnaires and 480 discussion-thread answers were considered as the sample.
Enhanced nutritional profile and follow-up, or improved physical activity practice	19.1 (9.1–33.3)	2 469 [9]	In addition, 349 answered questionnaires were considered as the sample.

The vast majority of studies covered the disciplines of maternal and sexual health, as well as gynaecology, mastology and reproductive health (40–42,46,47–54,59–70,72–80,82,84–86,88), but the included records also reported the impact and effectiveness of digital interventions on psychiatric disorders and mental health (41,45,56,61,71,78), oncology (81,83,87), endocrinology (46,55,82), surgery (63), emergency medicine (56) and physical well-being (46,55).

Thirty-seven records (RFO of 78.7% (CI 95%: 64.3–89.3)) emphasized how the use of DHTs would positively impact women's access to health services, allow an increase in self-care, and provide tailored self-monitoring systems to enable the acquisition of health-related interventions; this occurred via identifying appropriate health-care practitioners to provide services or locate practices, and by providing affordable, quality, comprehensive health care (Fig. 2) (40,47,48,51–60,62–65,67,68,70–82,83–88). The evidence showed that digital technologies enhance access rates to screening programmes, prompt lifestyle changes, and improve women's involvement in pertinent and relevant personal health matters (50,58,81–83,88).

Mathematical findings related to the impact of DHTs on women's health status are summarized in Fig. 2. As can be seen, 53.2% (95% CI: 38.1–67.9; 215 442 women from multiple countries and regions) of included primary studies stressed the positive impact of DHTs on maternal and gynaecological well-being and outcomes, either influencing women's pre-, intra- or postpartum stages or breastfeeding (40–42,46–54,59–70,72–80,82,84–86,88). Notably, digital technologies were reported by women as being as useful for seeking information about themselves or their baby (improving the longitudinal follow-up of gestation) (42,59,70,72,82), particularly in relation to their medication use during pregnancy (45,46,48,50,65,66,78,85,87), breastfeeding practices in low-, middle- and high-income countries and regions (66,70,76,81,83,85,87), and labour. Increased performance and access to mammography screening, treatment for irregular/intense menstrual bleeding, pregnancy-prevention advice and information about intrauterine devices, and a better understanding of childbirth were also positive outcomes of the use of DHTs.

According to Heitmann et al. (66), a Norwegian medicines information service for pregnant and breastfeeding women (the SafeMotherMedicine app) seemed to have fulfilled a previously unmet need to supply evidence on effective pharmacological interventions suitable during pregnancy and improved information about medications for pregnant and breastfeeding women in Norway. According to the authors, the products most frequently requested on an annual basis typically included medications for pain, nausea, yeast infection, haemorrhoids, allergies, asthma and mental health (66). Similarly, a cross-sectional online survey carried out with 749 users of mobile apps during labour in the United Kingdom in 2021 showed that women who used an app were significantly more likely to have gone into labour spontaneously, and those who delivered between 400/7 and 406/7 weeks' gestation were significantly more likely to use a labour-related mobile application ($X^2 = 12.68$; $P < 0.05$) (72). Most utilized phone applications that were used to monitor contractions, and most women found the apps useful and would recommend those they used (72). DHTs also offered emotional support during pregnancy and breastfeeding and were identified as a potential esteem-support mechanism for women.

The effect of health technologies utilized for the postpartum stage has been evaluated in six studies, from Bangladesh, Canada, Germany, Ghana, Switzerland and Türkiye (42,47,58,70,71,78). According to Perrenoud et al. (47), based on a Swiss socioanthropological study, immediate messaging applications (such as WhatsApp® and Viber®) had a positive effect on how postpartum women communicated with their providers. The applications were used by women to send questions, pictures and videos to health-care providers, allowing swift responses to their concerns (47). Midwives nevertheless encountered difficulties answering questions when they could not be solved through quick communication (for instance, in relation to infant crying).

Frequent use of messaging apps by parents may signify their need for emotional support, guidance or information related to parenting or personal matters. Recognizing this behaviour can provide health-care professionals with insights into parents' experiences, their comfort levels with technology-mediated communication and potential indicators of their well-being. Slomian et al. (40) found that women who used the Internet as a source of information believed the information was useful (288 of 349; 82.7%), positively impacting their postnatal period by serving as a so-called tranquilizer tool for specific infant symptoms and providing additional information on topics previously provided by health professionals.

Improvement of psychiatric-related outcomes (Fig. 2) was reported in 21 original studies (41,45,47,49,51,54,57,58,59,62,68,69,71–73,75,80,83,85,87,88), corresponding to an RFO of 44.7% (95% CI: 30.2–59.9), including patients from the United Kingdom (49,54,62,72,73,85,87) ($n = 7$; 14.8%), Germany (58,68) ($n = 2$; 4.2%), and Australia (83), Brazil (59), Canada (71), China (88), India (41), Japan (69), Kenya (51), New Zealand (80), Spain (45), Switzerland (47), the United States (57) and Viet Nam (75) ($n = 1$ (each); 2.1%). DHTs are associated with mental health improvement in multiple ways, including psychological relief and satisfaction after usage, mood improvement, women's reported sensation of happiness, fulfilment and pleasure, reduction in anxiety and overcoming of depression (41,45,47,49,51,54,57–59,62,68,69,71–73,75,80,83,85,87,88).

Fig. 2. Mathematical findings related to the impact of DHTs on women's health status



Primary studies also note how DHTs decrease workload and negative work-related patient experiences and offer emotional support to women. A reduced sense of isolation and feelings of loneliness, which some women felt unable to share with their families, followed DHT use, as did the garnering of a better body-image perception, highlighted as a modifying component of psychological-related outcomes. A small number of the included studies ($n = 5$; 10.6%) nevertheless associated the use of DHTs with negative emotional burdens (described as feelings of sadness, guilt, anxiety and being overwhelmed) and induction of psychological discomfort (46,49,55,56,61).

One RCT investigated whether a mobile app could impact the recording of nausea symptoms and vomiting during pregnancy, quality of life and decisional conflict regarding pharmacological interventions. Among the 268 women enrolled in the study, no evidence was found that the app could affect the tracking of the severity of nausea and vomiting symptoms (*adjusted* β 0.6, 95% CI: 0.1–1.2), quality of life (*adjusted* β –5.3, 95% CI: 12.5–1.9) or decisional conflict regarding pharmacological treatment (*adjusted* β –1.1, 95% CI: 6.2–4.2) compared with standard care. The analysis revealed that the changes in quality-of-life scores from baseline to the second week of the trial were not significantly different between the intervention and control groups (*adjusted* β –5.3, 95% CI: 12.5–1.9).

Less frequently reported outcomes are also displayed in Fig. 2. Enhanced nutritional profile and better dietetic follow-up with qualified health-care professionals, along with improvement of physical activity practice (RFO of 19.1% (95% CI: 9.1–33.3); $n = 9$; 19.1%) (40,45,46,55,59,76,78,82,88), better identification of life-threatening and severe pregnancy complications (RFO of 19.1% (95% CI: 9.1–33.3); $n = 9$; 19.1%) (40,42,52,62,70,73,75,78,86), and improvement in disease prevention and health promotion (RFO of 25.5% (95% CI: 13.9–40.3); $n = 12$; 25.5%) (41,48,50,58,59,61,67,71,75,76,78,88), are also among the most commonly rated topics in the included studies.

Three studies (6.3%; two from the United Kingdom and one from Germany) reported health outcomes associated with home-based medical termination of pregnancy (49,54,68). Overall, the use of DHTs positively impacted the care and health-related outcomes of women undergoing the process of pregnancy termination, either by providing access to a safe, legal and controlled-environment intervention, or by providing emotional relief for individuals throughout the difficult process. Boydell et al. (54) explain that patients significantly valued the option of accessing abortion care via telemedicine (as it provided high-quality health-care services) and emphasized the benefits of providing connection with health-care providers if needed. Women stated that the quality of abortion care was enhanced by the telemedicine service in relation to access, comfort and flexibility, and that ongoing telephone support also served as a psychological support tool for women who used the telemedicine abortion service (54).

Similarly, an online questionnaire study carried out in Ireland and United Kingdom (Northern Ireland) reported that women who completed at-home medical abortion through a telemedical platform commonly reported feeling “emotionally relieved” ($n = 716$ of 1023; 70%) and “satisfied” ($n = 368$ of 1023; 36%) with the process in general, along with almost half ($n = 501$ of 1023; 49%) reporting feeling “happy” and/or “pleased” (49).

A small proportion of patients reported negative outcomes, such as “feeling sad” ($n = 122$ of 1023; 12%), “guilty” ($n = 173$ of 1023; 17%) or “disappointed” ($n = 30$ of 1023; 3%) (49). These results align with the findings of Killinger et al. (68), who performed a content analysis of 255 emails following consultations from Germany in 2019 and highlighted the beneficial effects of abortion via telemedicine centres, which contributed to the development of a sense of well-being, construction of an emotional support network and reduction in delayed abortion approvals (68). Notably, all of the studies evaluating the applicability of DHTs to pregnancy termination were carried out in a domestic setting (patients’ homes).

Impact of DHTs on women’s empowerment and gender equality



Table 5 summarizes the meta-analysis of term- or content-based analysis regarding the use of DHTs on women's empowerment and gender equality.

Women's empowerment and gender equality primarily were achieved using DHTs to gain access to general resources (financial, informational, skills, services and social capital). Domain-specific display of the meta-analytical assessment is shown in Fig. 3.

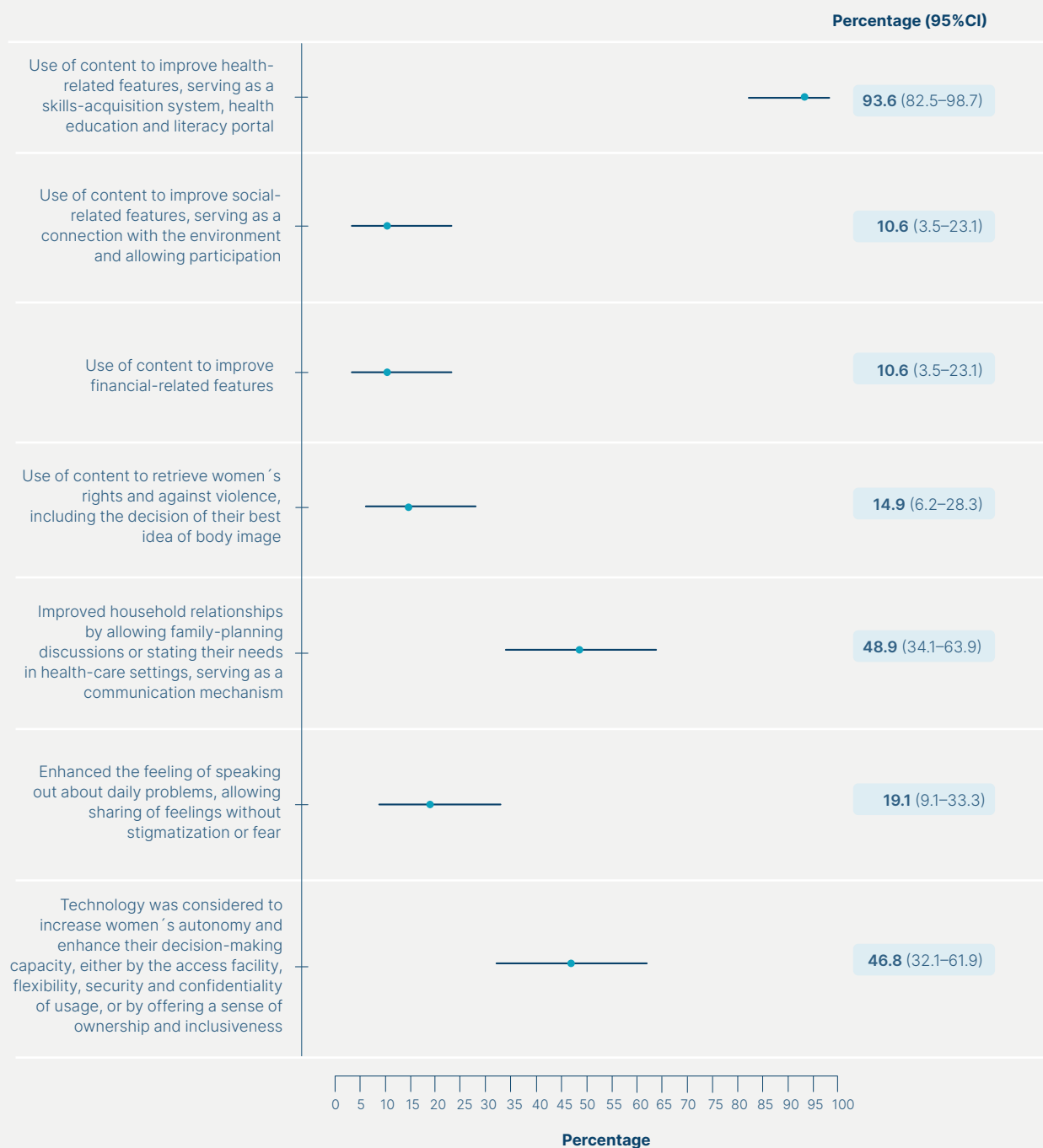
Overall, utilization of digital technologies directly impacted women's empowerment and gender equality either by serving as a skills-acquisition system or a health education portal/ health literacy platform (RFO of 93.6% (95% CI: 82.5–98.7); $n = 44$; 93.6%) (40,45–47,48–66,68,69,71–88). Primary studies reported that technologies were useful for health-related information acquisition, as contents improved understanding and engagement and served as a long-term encouragement mechanism for health-related issues (40,45–47,48–66,68,69,71–88). Use of DHTs was emphasized as a social-enabling tool because of its potential to connect women with society, increasing their communication with family members and health-care providers (RFO of 10.6% (95% CI: 3.5–23.1); $n = 5$; 10.6%) (40,41,45–47), or by improving financial situations (RFO of 10.6% (95% CI: 3.5–23.1); $n = 5$; 10.6%).

Financial-related outcomes were associated commonly with the use of DHTs that focused on enhancing access to affordable health-care services, reducing medical costs and increasing financial literacy. Financial-related planning resources empowered women through the acquisition of knowledge and tools required for better monetary decision-making related to health-care expenses and overall well-being, reduced women's labour workload and economic development of women. Women also reported that DHTs and their contents facilitated the retrieval of information about women's rights and actions against violence and improved their communication skills (family-centred or with health-care professionals) (RFO of 14.9% (95% CI: 6.2–28.3), $n = 7$; 14.8%).

Table 5. Summary of findings (empowerment and gender-related outcomes)

Outcomes (usefulness of DHTs)	RFO % (95% CI)	Number of participants [studies]	Comments
Use of content to improve health-related features, serving as a skills-acquisition system, health education and literacy portal	93.6 (82.5–98.7)	215 475 [44]	In addition, 480 discussion threads, 30 000 answered questions, 349 answered questionnaires and 501 posts were considered as the sample.
Use of content to improve social-related features, serving as a connection with the environment and allowing participation	10.6 (3.5–23.1)	66 200 [5]	In addition, 349 answered questionnaires were considered as the sample.
Use of content to improve financial-related features	10.6 (3.5–23.1)	68 160 [5]	–
Use of content to retrieve women's rights and against violence, including the decision on their best idea of body image	14.9 (6.2–28.3)	67 198 [7]	In addition, 349 answered questionnaires were considered as the sample.
Improved household relationships by allowing family-planning discussions or stating their needs in health-care settings, serving as a communication mechanism	48.9 (34.1–63.9)	203 601 [23]	In addition, 30 000 answered questions and 349 answered questionnaires were considered as the sample.
Enhanced the feeling of speaking out about daily problems, allowing sharing of feelings without stigmatization or fear	19.1 (9.1–33.3)	1 539 [9]	In addition, 349 answered questionnaires and 501 posts were considered as the sample.
Technology was considered to increase women's autonomy and enhance their decision-making capacity, either by the access facility, flexibility, security and confidentiality of usage, or by offering a sense of ownership and inclusiveness	46.8 (32.1–61.9)	42 132 [22]	In addition, 349 answered questionnaires and 501 posts were considered as the sample.

Fig. 3. RFO occurrence of findings associated with the impact of DHTs on empowerment and gender equity-related outcomes



The improvement of household and community relationships was demonstrated in 23 of the included primary studies (RFO of 48.9% (95% CI: 34.1–63.9); 48.9%) (Fig. 3), indicating that DHTs might be useful partner engagement systems and communication facilitators within the health-care apparatus (40,41,45–47,48,50,52,53,60,63,66,70–76,79,81,83,86). No study suggesting that using DHTs had negative results on household and community relationships was found. This domain was directly associated with the use of DHTs as communication mechanisms, as they frequently facilitate domestic discussions (including conversations on family planning) and empower women to state their needs to health-care providers. Open discussions enabled by DHTs that were promoted and performed domestically were described to extend beyond regular and standardized health-related interactions, encompassing crucial topics such as family planning, maternal health and general well-being. By providing a platform for domestic dialogue and involvement, DHTs empowered women to articulate their health-care needs more assertively and directly to health-care providers.

Technology was noted to increase women's autonomy and enhance their decision-making capacity in 22 records (RFO of 46.8 (95% CI: 32.1–61.9)), either through their easiness of use, flexibility, security and confidentiality of usage, or by offering a sense of ownership and inclusiveness (40,46,47,48,51–56,58–60,67,68,75–77,79–81,87).

The articles included in the review suggest that DHTs seem to offer a pathway for women to access information, support and health-care services in a more private, accessible and empowering manner, increasing their autonomy and decision-making capacity. Women commonly reported that they felt empowered to recognize, manage and explore appropriate management options for pregnancy-related symptoms without any boundaries posed by having physical mobility issues or without having to ask their partner's permission to access a health-care service or facility.

While the feeling of empowerment was recognized commonly by women, the perception may be more prevalent in certain cultural contexts or regions, including low- and middle-income countries. Women in several studies (40,46–48,51–56,58–60,67,68,75–77,79–81,87) reported that digital resources supported them to feel in control, helped to raise self-awareness and promoted self-management of symptoms, particularly during consultations with staff. One study related to pregnancy termination reported that digital technology helped women to feel comfortable and in control over the process during the consultation, with 20% ($n = 204$ of 1023) feeling confident and/or empowered using online telemedicine (49). Autonomy has been identified as the main internal motivation for choosing to use telemedicine for abortion, in addition to privacy and the desire to choose where, when and with whom to perform the abortion (68).

In two studies of the post-childbirth period, women reported feeling empowered by using the Internet, which allowed them to find out information on their own, check for information about specific baby symptoms and acquire additional information (47,78). One study found that a web-based service empowered women to make informed decisions regarding medication use during pregnancy and breastfeeding (66). Another that focused on breastfeeding reported that having access to online support groups provided women with self-identity, feelings of worth and external validation (85).



Factors affecting access and use of DHTs

Table 6 indicates the categories of barrier identified among the included studies, followed by examples highlighted among primary studies.

Though modes of women's empowerment via DHTs were evident, significant barriers to access and utilization continue to limit the applicability of these health technologies to users, particularly those excluded due to the ongoing digital divide. Socioeconomic factors (including family income, education, employment status and age) proved relevant to the applicability of these health technologies.

A framework developed for the review identified 10 factors that limit female patients' access to DHTs worldwide (Fig. 4 and 5).

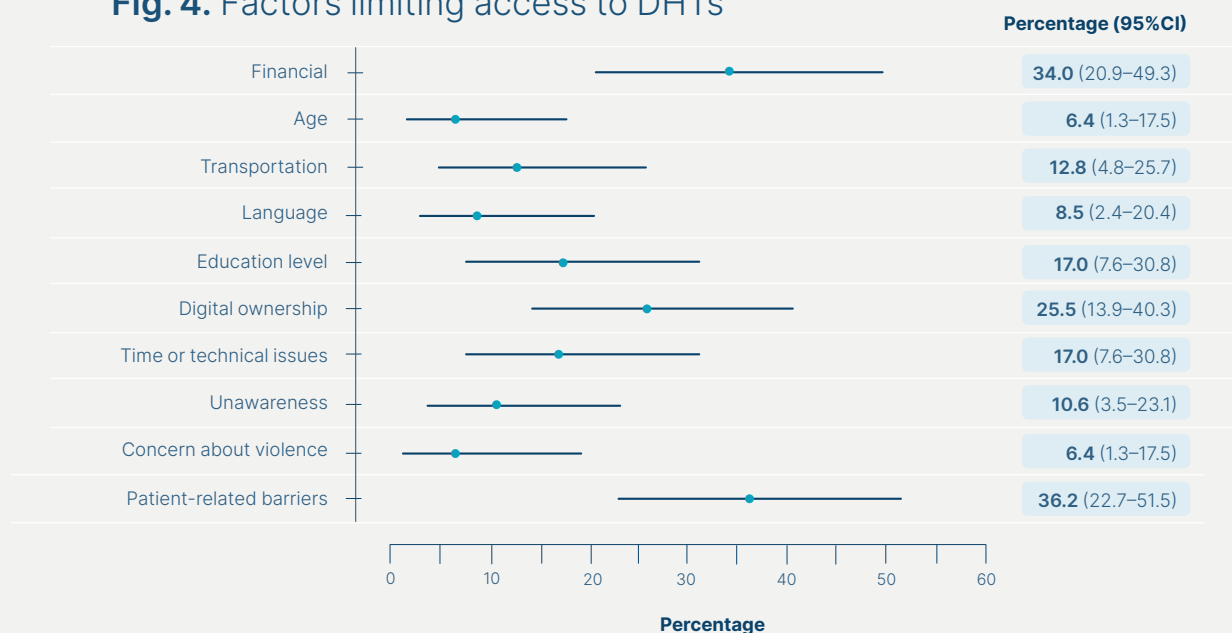
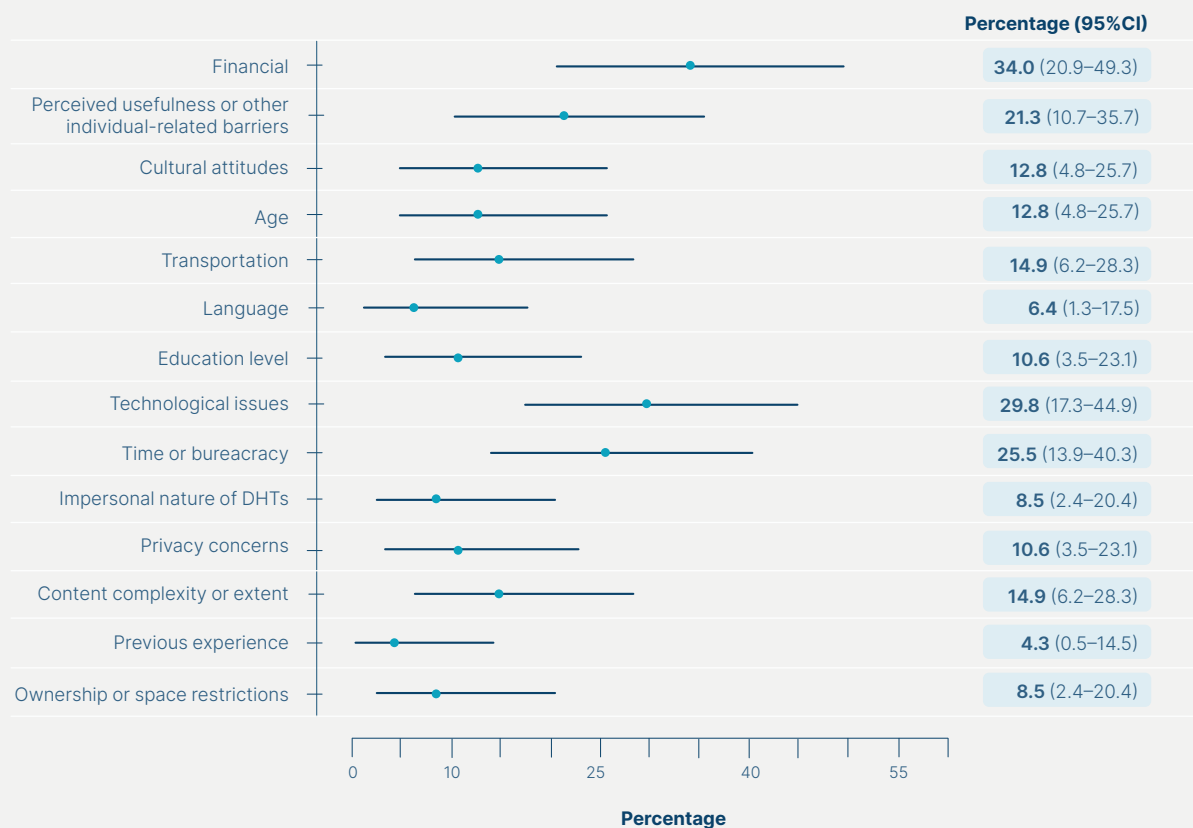
Financial barriers (limited monetary resources for purchasing devices and limited funds for the payment of phone carriers' plans) were observed in 11 studies (RFO of 34.0% (95% CI: 20.9–49.3); 23.4%) (48–50,53,70,71,73,76,79,86,88). For example, users were unable to afford to pay for messages/calls received or missed, which prevented initial or continued access to the health technology platform.

The most common barrier for access to DHTs was patient-related (RFO of 36.2% (95% CI: 22.7–51.5); 36.1%), which was relevant to 16 of the included studies (41,45,52–54,56,58,63,68,70,72,80–83,86,88). These barriers encompassed issues such as personal unfamiliarity with online care, partner-related barriers (such as the denial of phone access by male partners), personal issues and other familial/cultural factors.

Digital ownership offered a third prominent barrier to DHT access (RFO of 25.5% (95% CI: 13.9–40.3)) (41,42,48,53,70,75,76,79–81,83,88). Women who did not have access to a phone, smartphone or computer were unable to access the health services provided by DHTs. These modes of DHT delivery were largely impacted by factors such as socioeconomic status or the need for permission/provision from male significant others, with financial and digital-ownership barriers comprising two of the most prominent barriers to access. Gender inequality therefore permeates the ownership of technology from the start, which serves as a collective challenge to the usability and generalizability of DHTs for women.

Table 6. Categories of barriers identified in included studies with examples from primary studies

Categories of barriers	Examples from primary studies
Network	Limited Internet connection, expensive data plans, lack of subscriptions to plans, low-quality signal, and inadequate or unreliable Internet connectivity in rural or remote areas
Accessibility	A particular focus for people with disabilities and older women, lack of accessible technology devices (such as smartphones and computers) due to affordability issues, and physical disabilities affecting the usability of digital tools (particularly those lacking accessibility features)
Utility	Lack of relatability and lack of information on recommendation use, poor user interface design or language barriers in digital health apps limiting usability, and limited availability of localized or culturally relevant content in digital health platforms
Digital ownership	Having an individual device for multiple purposes and limited control or access to shared digital devices within households
Policy change	Implementation of certain technologies requiring systematic change in management, regulation and implementation processes, lack of supportive policies or regulations promoting equitable access to digital health services, and legal barriers or restrictions limiting women's access to certain health-related information or services online
Gender divide in education	Restricted access to educational systems, minimal incentives for women to study, gender disparities in digital literacy and education affecting women's ability effectively to use digital health tools, and cultural or societal norms limiting women's opportunities for education in technology
Individual patient factors	Digital literacy, perceived usefulness, patience for utilization, lack of digital skills, lack of awareness or knowledge about available digital health resources and their benefits, and reluctance or resistance to adopt digital technologies due to cultural beliefs or preferences for traditional health-care practices
Geographic	Distant municipalities or difficult access to cities through conventional means of transportation and urban–rural disparities in access to digital health services and resources
Economic/financial	Being able to pay out of own pocket for devices and their associated running costs, financial constraints limiting women's ability to afford Internet data plans or necessary devices, and costs associated with digital health services or apps creating barriers for low-income women
Socioeconomic	Lack of time to use the device due to the burden of household activities or professional workloads, lack of household support, socioeconomic disparities influencing women's access to education and employment opportunities affecting digital literacy, and unequal distribution of resources and infrastructure in underserved communities impacting on access to DHTs

Fig. 4. Factors limiting access to DHTs**Fig. 5.** Factors limiting the utilization of DHTs

Fourteen barriers to DHT utilization were also recognized within the data-extraction process. The most prominent factor affecting utilization was found to be financial barriers (RFO of 34.0% (95% CI: 20.9–49.3)), elicited in 16 of the included studies (42,45,48–50, 52,53,68,70,73,76,79,84,86,88,89). Financial barriers therefore presented significant hindrances to both DHT access and use, a reality that is pertinent to potential future mitigation strategies.

Technological factors and time/bureaucratic issues offered two similarly prominent barriers to DHT utilization, with RFOs of 29.8% (95% CI: 17.3–44.9) and 25.5% (95% CI: 13.9–40.3) respectively. Technological issues ranged from problems with connecting to or maintaining network connections and application glitches. The issue of network errors proved very common, as many of the explored DHT interventions occurred in underresourced areas/countries in which access is perhaps most needed, yet the most difficult to attain.

The barrier of time proved relevant to gender roles, as time spent on the DHT intervention frequently conflicted with women's professional, marital and maternal roles. As is observed commonly in societies, women are challenged with multiple tasks and responsibilities beyond their own health concerns and needs, including professional commitments, household responsibilities and caregiving functions. Consequently, having sufficient time to engage with and utilize DHTs is still considered a constant variable that women might overcome. Time for utilizing and implementing DHTs did not prove feasible given the additional multifaceted responsibilities carried by women outside of their duties toward personal health.

A final barrier toward DHTs utilization was perceived usefulness or other individual-related barriers, with a relative frequency occurrence of up to 21.3% (95% CI: 10.7–35.7). This barrier highlights the need for education surrounding the effectiveness and importance of these technologies for end-users. Individual barriers to utilization were mirrored in the access stages, eliciting the need for an individualized methodology of DHT implementation to improve both access and usability for women on a global scale. All meta-analytical assessments of barriers to access and utilization of DHTs reported by women among included studies are shown in Tables 7 and 8.



Table 7. Summary of findings (barriers to access identified in included articles)

Factors	RFO % (95% CI)	Number of participants [studies]
Financial	34.0 (20.9–49.3)	212 478 [16]
Age	6.4 (1.3–17.5)	3 100 [3]
Transportation	12.8 (4.8–25.7)	68 389 [6]
Language	8.5 (2.4–20.4)	1 420 [4]
Education level	17.0 (7.6–30.8)	202 175 [8]
Digital ownership	25.5 (13.9–40.3)	114 889 [12]
Time or technical issues	17.0 (7.6–30.8)	9 261 [8]
Lack of awareness	10.6 (3.5–23.1)	160 934 [5]
Concern about violence	6.4 (1.3–17.5)	1 157 [3]
Patient-related barriers (personal barriers)	36.2 (22.7–51.5)	71 143 [17]

Table 8. Summary of findings (barriers to utilization identified in included articles)

Factors	RFO % (95% CI)	Number of participants [studies]	Comments
Financial	34.0 (20.9–49.3)	146 520 [16]	–
Age	12.8 (4.8–25.7)	42 552 [6]	–
Cultural attitudes	12.8 (4.8–25.7)	41 310 [6]	–
Language	6.4 (1.3–17.5)	1 300 [3]	–
Education level	10.6 (3.5–23.1)	41 841 [5]	–
Transportation	14.9 (6.2–28.3)	2 998 [7]	–
Time or bureaucracy	25.5 (13.9–40.3)	10 946 [12]	–
Technological issues	29.8 (17.3–44.9)	4 616 [14]	In addition, 30 000 answered questions were considered as the sample.
Perceived usefulness or other individual-related barriers	21.3 (10.7–35.7)	3 688 [10]	In addition, 349 answered questionnaires and 501 posts were considered as the sample.
Impersonality	8.5 (2.4–20.4)	548 [4]	In addition, 30 000 answered questions were considered as the sample.
Privacy concerns	10.6 (3.5–23.1)	2 477 [5]	–
Content complexity or extent	14.9 (6.2–28.3)	8 450 [7]	In addition, 30 000 answered questions were considered as the sample.
Previous experience	4.3 (0.5–14.5)	614 [2]	–



Discussion

This systematic synopsis of the current evidence and content-based meta-analysis provides a comprehensive evaluation of the scientific literature on the association between digital technologies and women's health promotion, achievement of equality and empowerment from a global perspective.

The findings suggest that various technological modalities within DHTs have shown promising uptake and utilization for improving certain aspects of women's health status. While these interventions demonstrate potential in contributing to the enhancement of women's health outcomes, it is important to note that their extent and effectiveness may vary across different contexts and settings.

The assessments suggest that overall, DHTs positively improve outcomes related to gynaecology, obstetrics and psychiatry. They also have a relevant influence in enhancing nutritional intake and longitudinal follow-up of women (40,45,46,55,59,76,78,82,88), and promoting physical activity. Notably, digital interventions play a crucial function in empowering women by giving them the opportunity to obtain access to resources on their own (health-, financial-, personal- or social-related information), offering them an array of digital opportunities, and strengthen women's communication skills with partners, family members and health-care providers (40,41,45–48,50,52,53,60,63,66,70–76,79,81,83,87). In general, the findings show that by using DHTs, women significantly increase their autonomy and enhance their decision-making capacity, as they allow them the possibility of deciding plans of action by themselves in a flexible, safe and confidential environment.

Gender divide and digital gender gap



The review is timely and relevant since the most recent statistics on the gender divide are worrying. According to the International Telecommunication Union, the gender parity score between 2019 and 2022 increased by only 0.02 points on a scale ranging from 0 to 1 (the closer to 1, the higher gender parity is achieved) (27). In fact, the digital gender gap between these two time points actually increased by 20 million, as the number refers only to the ratio of percentages. A score of 1 has been reached only in the Americas and the Small Island Developing States (27).

Several global-scale initiatives have recently been undertaken to tackle digital gender divide discrepancies, including the WHO universal health coverage guidelines, which apply an

intersectional gender lens to health-care plans of action to ensure that women and girls who are disproportionately financially disadvantaged are not left behind during the evolution process of medicine and technology (90,91).

Despite the ongoing rapid technological revolution, however, low- and middle-income countries still face challenges in offering equitable and supportive conditions to secure women's access to digital technologies (92). The relatively low numbers of studies from African, Asian, Caribbean and Latin American countries and regions observed by the review may reflect reduced focus on this topic in society and academia, indicating the need for more research in the field.

To reduce access inequalities and provide universal and affordable access to technology systems in the least developed countries, as endorsed by the SDGs, unprecedented efforts at global scale should remain the focus of multiple social agents, including civil societies, philanthropic organizations, policy-makers and corporations. Researchers, health-care providers and policy-makers should consider the representativeness of countries actively evaluating the fields of “Digital Health” and “Women's Health”, as the results demonstrate a predominance of studies from high-income settings.

DHT potential in medical specialties



Most of the included studies covered the broad fields of gynaecology and obstetrics, reproductive health and sexual health and their association with digital health and gender, equality and women's empowerment. The findings suggest that digital technologies are positively beneficial to women's health-related outcomes, as they increase access to specialized services in multiple medical specialties, leading to higher performance levels, and promote access to screening tests and pharmacological and nonpharmacological interventions. DHTs also serve as a positive modifier to the increased utilization of screening services and act as a platform from which to retrieve disease prevention advice and information about medical issues and conditions, particularly among socially vulnerable women.

DHTs, however, have potential applicability well beyond this relatively narrow range of medical specialties, as highlighted in a short communication by Harman in 2022 (93). Various clinical trials suggest that DHTs might be useful in the fields of dermatology, psychiatry, rheumatology, and cardiology, particularly to women patients (94–97). DHTs can also be useful in evaluating workers' health, including unpaid or informal carers.

Pregnancy termination

Some studies proposed the positive perception of DHTs associated with pregnancy termination (49,54,68). Utilization of DHTs in home-based medical termination of pregnancy may offer women a sense of enhanced privacy, allowing them to undergo a sensitive medical intervention in a more discrete and comfortable environment (98,99). This privacy aspect might be particularly crucial in regions or communities in which stigma or social barriers surrounding abortion access persist (100).

Exploring the issue of accessibility is imperative to understanding the broader impact of DHTs in this context. Accessibility encompasses not only the physical availability of medical services, but also factors such as geographic location, socioeconomic status and legal restrictions (101). Digital solutions could potentially address barriers to access by reaching women in remote or underserved areas where traditional health-care services might be limited or inaccessible. Such technologies might offer a lifeline to women experiencing various difficulties accessing abortions, as they provide a safe and confidential alternative. The positive impact of DHTs on health outcomes during home-based medical termination of pregnancy therefore extends beyond medical safety and emotional relief by potentially enhancing privacy and addressing accessibility barriers for women seeking these services.

Forms of DHTs



The inclusion and exclusion criteria on type of DHTs to include in the study were broad. Although multiple forms of digital interventions were included, only a limited variety of modalities was retrieved from included primary studies.

Mobile health technologies (41,42,45,47,48,51–53,57,59–61,64–66,69,70,72–81,83,86,88) were cited more frequently than telehealth and telemedicine interventions (40,41,46,49,50,54,56, 58,63,67,68,71,84) and social media platforms (41,55,62,85,87). The findings align with recently reported data on the use of smartphone-based applications for the improvement of multi-illness conditions (102–105).

Other modalities of DHTs besides mobile health platforms may nevertheless have an extremely important role in women's health. The Centers for Disease Control and Prevention stress that more than 37% of adults in the United States used telemedical platforms and channels in 2022, with higher usage rates among women (42.0% versus 37.0% of male users), non-Hispanic citizens (39.2% versus 32.8% of Hispanic consumers) and more educated individuals (43.2% among adults with a college degree or higher versus 28.7% among adults with less than a high-school diploma or General Educational Development certification) (106).

Similarly, findings from the statistical office of the European Union (Eurostat) show that women used the Internet more than men for seeking health information (68% of women and 56% of men), making phone or video calls (75% and 71%), and engaging in social networks (66% and 62%) (107). Technologies involving the use of Internet of Things concepts and machine-learning techniques have risen rapidly in relation to obesity management, pregnancy development, gynaecological and endocrinological calculations and performance evaluations, and oncology (108–112). Future scientific evaluations should therefore focus on the use of various types of DHTs for improving women's health-related outcomes, as promising results have been observed over recent decades.

Empowerment and equality



The review found evidence that women gain increased power and control over their own lives and acquire the ability to take strategic decisions by obtaining useful information through DHTs (either health- or financial-related, or other types), and increase their sense of social connectivity (40,42,45–66,68,69,71–88). The empowerment of women through the utilization of DHTs encompasses various dimensions beyond the acquisition of information and social connectivity (40,41,45–47).

The review finds that access to diverse digital resources, ranging from health-related information to community participation and beyond, equips women with valuable tools and knowledge to make informed decisions concerning their health, finances and overall well-being. The ability to make strategic decisions based on their own needs, life experiences and expectations (which might particularly be tailored to subgroups of women) is found to represent an enhancement to women's agency and autonomy.

DHTs serve as catalysts in fostering a sense of autonomy among women, actively engaging them in decision-making processes related, but not limited, to treatment options, health management, preventive measures and lifestyle choices. The empowerment gained through autonomy extends to increasing access to information and enhancing women's personal autonomy to define their own life-courses and future wishes. The empowerment experienced by women through digital technologies is not a one-dimensional outcome, but an ongoing process influenced by various factors such as digital literacy and accessibility, and socioeconomic contexts. Fostering digital inclusion and ensuring equitable access to technologies has become imperative to realizing and sustaining women's empowerment in diverse communities and settings.

Merging women's empowerment and gender equality into medical research projects requires interdisciplinary collaboration and a robust theoretical framework. Multispecialty research enables the combination of perspectives, expertise, academic and professional backgrounds that allows a comprehensive understanding of factors that might modulate women's health and their experiences with DHTs. The engagement of researchers from multiple fields of expertise ensures that primary studies are designed and interpreted as a unique phenomenon, based on

inclusiveness, representativeness and legitimacy principles (113). A solid theoretical framework functions as a tool for understanding and contextualizing research findings in relation to women's empowerment and gender equality (114). The utilization of social theories, intersectionality and social determinants of health into a theoretical framework provides valuable perspectives on the variables influencing women's health disparities and inequities (115).

While these concepts serve as core guiding elements, consideration of context fluidity and the content-dependent nature is of the utmost importance. The interpretation of research findings within the interface between women's empowerment and gender equality often varies significantly according to the ideological perspectives of the personnel conducting and coordinating the research (a clinical trial assessing the importance of telemedicine and telehealth services for pregnancy termination, for instance, might be drafted and reported differently if primarily conducted by male researchers) (116,117).

Women's empowerment and gender equality has evolved, extending from the simplistic individual autonomy point of view to cover the intrinsic link between individual transformation and substantial structural societal shifts. It is timely to emphasize that these concepts are no longer confined merely to personal-oriented perspectives, but also take into consideration the necessity for wider societal transformations to adequately empower women within societies.

Barriers to access and utilization



The results of this meta-analysis indicate numerous multifaceted barriers to access and utilization of DHTs by women. Ownership (or lack of it) of a digital modality, such as a phone, smartphone or computer, presents a primary barrier to the use of technological platforms. Financial barriers proved widespread across the studies, affecting both initial access and continued utilization of DHTs, a factor relevant to the possibility of digital ownership. Additionally, increased socioeconomic status and dismantling of social strata improve access to digital technologies.

While gaps in digital ownership might appear to originate from socioeconomic disparity – where resourced, being affluent and being educated were associated with higher access to technology – the power imbalances and gender inequity felt by many women permeated this barrier. For example, having permission to own a phone or smartphone, often from a male partner, was highlighted by some women as being essential to achieving access. Women also noted having to use their husband's phone for certain digital interventions, which both decreased usage and hindered access.

The fact that gender inequities influenced the use of DHTs by women in households that already had access to technology indicates that empowerment should be prioritized. Data from the United Nations University Equals Global Partnership showed significant disparities in digital use between women and men across the globe (118). When the level of IT was controlled for, even developed and technologically advanced nations such as Japan and the Republic of Korea

exhibited significant disparities in digital use (118). The digital disparity is actually increasing: 250 million fewer women were online in 2017 compared to 2013 (118). Socioeconomic relief therefore is not the sole solution to this limit on digital access, as gender inequity transcends socioeconomic barriers.

Socioeconomic status, financial power and gender equality imbalances are not the only factors influencing digital ownership and usage. The mobile gender gap report of the Global System for Mobile Communications Association in 2018 highlighted how in an analysis of 40 global economies, all countries (except Burundi) reported an at least 50% rate of women owning a mobile phone (119). This indicates that simple ownership and direct access to a digital technology modality is not the root cause of these disparities in utilization.

While the review's findings suggest disparities in utilization of digital technology modalities, it is essential to acknowledge that universal access to technologies such as mobile phones may not be consistent across all populations (119,120). Disparities in utilization may stem from various factors beyond mere ownership or direct access to digital technology (119,120). Barriers such as limited infrastructure, geographic constraints, economic disparities and varying levels of digital literacy may significantly impact the utilization of these technologies among certain demographic groups or in specific regions (119,120).

Even women who have access to technologies can experience limited utilization. As identified above, the most common barriers to utilization were cited as technological issues, time and perceived usefulness of the intervention. While technological issues prove relevant to the geographical isolation of countries and regions and necessary increases in technological expense to developing countries, time and perceived utility impact at a more individual level.

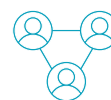
The reason cited for the inability to dedicate time to a personal digital health intervention is often the female user's conflict with multifaceted additional roles, including familial, maternal and professional/job-related duties (118). These significantly impede female users' ability to dedicate time to an intervention that would aid their personal health status. Women work fewer hours in paid environments and many more hours on unpaid, domestic duties compared to men (118). This pattern holds true for every single analysed country across Africa, the Americas, Asia, Europe and Oceania within this United Nations data framework (118).

The work-life balance barrier presents an area in which improvement is needed in relation to gender equality in employment and professional autonomy, with dedicated female users not only being the custodians of their own personal health, but also being mothers, workers and partners. Until equality is prioritized in these additional areas, personal health-related matters will continue to be neglected.

The language around women's empowerment and promotion of DHTs to enable women's health-care access and management is imperative. The digital divide involving women must not be evaluated as a series of solo elements. Financial access, national economies and professional capability are certainly critical components when addressing digital disparities and existing social, economic and cultural inequalities, but human rights issues must be also considered to promote full participation of women in society.

Preserving the fundamental right to health care and ensuring equitable access to digital technologies for women are pressing and relevant issues in the digital age. The empowerment of female DHT users and women in general is inevitably multidimensional in nature, requiring socioeconomic support, financial equality, professional equity and humanitarian prioritization to mitigate the abundant barriers at play. An adequate framework to address these barriers to access and utilization of DHTs by women worldwide must be multifaceted, requiring institutional support, initiatives by nongovernmental and multinational organizations, professional and personal aid, educational reform and socioeconomic relief.

Potential actions for stakeholders to optimize the effects of DHTs in practice



Box 2 identifies some potential actions that could be useful in addressing the barriers observed in the included studies. It suggests that digital infrastructure should be expanded to ameliorate issues of access and ownership for female users. This includes physical expansion to remote or rural areas often isolated from network access, and expanded service provision that is more attuned to the current needs of women, with greater working capacity.

More data relative to women's access and the digital gender divide are needed. Affordability measures can be developed to enable women, regardless of socioeconomic status, to have equitable access to DHTs. Equality targets should be prespecified and digital safety movements pursued.

Digital literacy needs to be improved to reflect the requirements of an increasingly connected world. The digital gender divide will continue to expand while those in greatest need of DHT services remain disconnected and excluded. While digital literacy is redefined continuously, it can be addressed through education initiatives prioritizing both female reading literacy and digital literacy via educational reform and national initiatives. Policy-makers should highlight the need for legislation pursuing gender equality in education. Community-based interventions are often cited as successful initiatives in this realm, as is the further promotion of accessible education to narrow the digital health divide. These kinds of digital policies can be supported by the work of targeted nongovernmental organizations (NGOs), national initiatives and organizations such as the Broadband Commission.

Socioeconomic and cultural factors need to be considered if barriers to DHT use related to the persistent gender disadvantage felt by women are to be addressed. Relevant stakeholders include academia, policy-makers, NGOs and organizations focusing on community interventions.

Personal barriers related to the pressure felt by many women to maintain numerous roles (familial and professional) should be brought into focus. Measures to address the gender pay gap could be prioritized to mitigate the work, time and monetary resources cited as barriers to female digital use.

The only way to effect global change is via truly global dialogue. This is a pressing issue, but not one that can be resolved by a single entity. Collaborative and concerted efforts targeting change and specifying gender equality within the realm of DHTs must involve institutional, organizational and educational reparations. Only then can a truly connected digital world be achieved, and gender eliminated as a predictor of health-care access and outcomes.

Box. 2. Potential actions for stakeholders

To address the barriers observed in the included studies to women's access to, and utilization of, DHTs, stakeholders could consider the following.

Boosting infrastructure, education and community-based service provision by:

- ▶ expanding digital infrastructure capacity and reach;
- ▶ expanding service provision that is more attuned to the current needs of women;
- ▶ launching digital literacy improvement initiatives;
- ▶ promoting community-based educational interventions; and
- ▶ introducing educational reform and gender equity targets.

Improving social awareness and enhancing data collection and evidence generation by:

- ▶ obtaining more data related to female access to technology and the digital health divide;
- ▶ implementing media campaigns showing the effectiveness of multiple technologies in improving women-related outcomes; and
- ▶ launching a community-based movement to support DHT use.

Enhancing policy design and sustainable community outcomes by:

- ▶ creating social programmes and campaigns to enable women to acquire their devices at a lower rate;
- ▶ highlighting the need for legislation to pursue gender equity in education;
- ▶ calling for global dialogue on institutional, organizational and educational reparations;
- ▶ tracking the gender pay gap and providing socioeconomic relief;
- ▶ facilitating the consideration of socioeconomic and cultural factors by relevant stakeholders, including academia, policy-makers, NGOs and organizations focusing on community interventions; and
- ▶ providing incentives to stimulate modern grid investments that offer societal benefits.

Limitations



The research simultaneously analyses the impacts and effects of DHTs on women's health status and on improving gender equality and empowerment globally. Well known quality-related benchmarks in the field of evidence-based medicine were used, including prior publication of the project management plan (in the PROSPERO platform), rigorous and systematic screening of studies and their assessment, and data extraction and reporting. A wide range of sources, hand-searched studies primarily deemed eligible for inclusion and grey literature from reputable health sources was reviewed.

Some limitations associated with the project nevertheless bear mention. First, the utilized technique for data summarization (thematic analysis) involves biases associated with the style of writing of the primary authors and their interpretation of participants' data and given opinions, particularly in studies where transcripts were taken up and used. This element, related to the reporter's objectivity, might subconsciously be reflected in final findings. By being aware of this limitation, however, the review team tried to involve researchers from different academic backgrounds and areas of expertise to evaluate the obtained data in a less biased way.

Secondly, it was decided to limit eligible studies to original primary data. Expert letters, editorials, correspondence and opinion-based publications, which are acknowledged as legitimate providers of research data, were therefore excluded (121,122). The rationale for this criterion is based on several results stating the higher likelihood of these types of publication having industry ties than supportive articles, and the elevated reporting and selection bias they feature (121,122). It was decided not to use a formal quality-appraisal tool, as the quality of the evidence itself was not the primary target, but the concepts addressed.

Lastly, the outcomes were limited by the heterogeneity and quality of the evidence expected by the panel of experts in our research team using a defined data-extraction framework. This partially narrowed the assessment foci but was counterbalanced by the depth concepts raised in the approach.



Conclusion

A global scoping review of digital health and women's health, including empowerment and gender equality, was conducted. The study identified improvement in health-related outcomes associated with maternal and sexual health, gynaecology, mastology and reproductive health. Evidence on the effectiveness of digital interventions in psychiatric disorders, mental health, oncology, endocrinology, surgery, emergency medicine and physical well-being was collated, demonstrating that DHTs also facilitate self-care practices and support tailored self-monitoring systems, including increased access to screening programmes, lifestyle changes and enhanced involvement of women in personal health domains.

By summarizing the existing literature from qualitative and quantitative original studies, the review was able to identify numerous barriers to accessing and utilizing DHTs that women worldwide face. It pinpointed the absence of scientific reports on certain modalities of digital interventions, in addition to the targeted focus of these technologies in gynaecology and obstetrics, and in high-income countries.

The review emphasized the immediate call for action for multiple social actors to address a broad range of barriers identified throughout the research. Based on a multidisciplinary workforce, initiatives to improve existing digital infrastructure, enhance digital literacy and promote gender equality through education are fundamental to overcoming most of the identified barriers. Legislative and regulatory reformation and international dialogue among competent agencies and institutions are equally relevant in bridging the digital divide and gender disparities, ultimately unlocking the potential of DHTs in improving women's health and advancing gender equality.

The methodology of thematic analysis allowed a determination to be made of the main domains in which DHTs positively influence the establishment of women's empowerment and gender equality, based on the relative frequency of occurrence. The study has pertinent formal implications for research academics and policy-makers, especially in view of the current transformative digital age and truly global health network.

Regarding the research perspective, the literature assessment provides guidance for a practical strategy to consider when designing globally applicable original studies. From a policy point of view, the study suggests that future guidelines should focus on how best to enrol women in research concerning the use of digital interventions on health-related outcomes while prioritizing access, utilization and barrier mitigation. The approach should be collaborative and should involve a wide range of stakeholders, including medical providers, social sciences specialists, politicians, ethical review committee members and patients. In this vein, the pertinence of women's health, positive health outcomes and equitable health-care access within the digital sphere can be adequately targeted for future interventions to address the ever-increasing digital divide.



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Annex 1. Terminology used in the project

The purpose of this annex is to define a basic terminology system relating to the concepts introduced throughout the report and annexes.

Artificial intelligence – refers to systems designed by humans that, given a complex goal, act in the physical or digital world by perceiving their environment, interpreting the collected structured or unstructured data, reasoning on the knowledge derived from these data and deciding the best action(s) to take (according to pre-defined parameters) to achieve the given goal.

Big data – in health, big data refers to large routinely or automatically collected datasets that are electronically captured and stored. The data are reusable in the sense that they are multipurpose. It involves the fusion and connection of existing databases for the purpose of improving health and health system performance. It does not refer to data collected for a specific study.

Deep learning – a particularly accurate machine-learning approach with less need of human guidance, referring to the fact that the neural network has several layers between the input and the output, learning the overall input– output relation in successive steps.

Digital health technologies – refers to the use of information and communications technologies in medicine and other health professions to manage illnesses and health risks and to promote wellness. Digital health has a broad scope and includes the use of wearable devices, mobile health, telehealth, health information technology and telemedicine. Digital health has been gaining momentum because it is envisioned to: improve access to health care; reduce inefficiencies in the health-care system; improve the quality of care; lower the cost of health-care; and provide more personalized health care for patients.

eHealth – is defined as digital health and care referring to tools and services that use information and communication technologies to improve prevention, diagnosis, treatment, monitoring and management of health-related issues and to monitor and manage lifestyle habits that have an impact on health.

Electronic medical record – digital versions of the paper charts in clinician offices, clinics, and hospitals. They contain notes and information collected by and for clinicians in the office, clinic or hospital and are used mostly by health-care providers for diagnosis and treatment.

Empowerment roadmap – addresses the third priority of digital transformation of health and care and outlines what elements are needed, in particular at organizational level, to improve patients' empowerment and therefore promote better collaboration with health-care providers and health systems while increasing their trust in digital health.

Health technology – a medicinal product, medical device, or medical and surgical procedures, as well as measures for disease prevention, diagnosis or treatment used in health care.

Internet of Things – a dynamic global network infrastructure with self-configuring capabilities based on standard and interoperable communication protocols where physical and virtual things have identities, physical attributes and virtual personalities, use intelligent interfaces, and are seamlessly integrated into the information network.

Machine learning – a subdiscipline of artificial intelligence, denoting the ability of a piece of software to learn from its environment or from a very large set of representative data, enabling systems to adapt their behaviour to changing circumstances or to perform tasks for which they have not explicitly been programmed.

Personalized medicine – refers to a medical model using the characterization of individuals' phenotypes and genotypes (for example, molecular profiling, medical imaging and lifestyle data) for tailoring the right therapeutic strategy for the right person at the right time, and/or to determine the predisposition to disease and/or to deliver timely and targeted prevention. Personalized medicine relates to the broader concept of patient-centred care, which takes into account that in general, health-care systems need to respond better to patient needs.

Telehealth – a broader definition than telemedicine, as it includes computer-assisted telecommunications to support management, surveillance, literature and access to medical knowledge.

Telemedicine – the provision of health-care services through the use of information and communications technology in situations where the health professional and the patient (or two health professionals) are not in the same location. It involves the secure transmission of medical data and information through text, sound, images or other forms needed for the prevention, diagnosis, treatment and follow-up of patients.

Women's empowerment – a multidimensional social process that challenges unequal power relations by which women gain the ability to exercise control over their lives, to make strategic life choices and to shape systems around them, including in the economic, sociocultural, family/interpersonal, legal, political and psychological realms.

Women's health – refers to women's state of complete physical, mental and social well-being. It is also understood as the branch of medicine that focuses on the treatment and diagnosis of diseases and conditions that affect women's physical and emotional well-being. It is commonly associated with sexual and reproductive health but also includes all aspects of the health of women related to their biology and to gender-related differences that may impact on access and use of health services and resources.



Annex 2. Search strategy

Excerpta Medica Database (Embase)

Embase Classic + Embase 1947 to 2023 August 18

Embase Classic + Embase <1947 to 2023 August 18>

- 1 exp digital divide/
- 2 (digital adj3 (divide* or equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or diversity or balance* or unbalance* or imbalance*)).ti,ab.
- 3 1 or 2
- 4 exp female/or exp "gender and sex"/or exp transgender/or exp pregnancy/
- 5 (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*).ti,ab.
- 6 4 or 5
- 7 3 and 6
- 8 gender equity/or women's rights/
- 9 ((female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) adj3 (equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or discrimination* or disaggregat* or diversity or balance* or unbalance* or imbalance* or right*)).ti,ab.
- 10 ((female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) adj3 (use* or usage* or utili#ation or access*)).ti,ab.
- 11 unequal world.ti,ab.
- 12 8 or 9 or 10 or 11
- 13 exp Digital Technology/or exp Information Technology/
- 14 (digital adj3 (technolog* or electronic* or environment* or age or inclusion or era or world)).ti,ab.
- 15 ((information or communication) adj3 technolog*).ti,ab.
- 16 (knowledge society or digiti#ation).ti,ab.
- 17 exp telehealth/
- 18 ((tele* or digital or mobile or smart) adj3 health).ti,ab.
- 19 (telehealth or telemedicine or telemonitoring).ti,ab.
- 20 exp mobile application/
- 21 internet/

- 22 exp mobile phone/
- 23 computer assisted therapy/
- 24 personal digital assistant/
- 25 (app or apps).ti,ab.
- 26 (online or internet or web* or digital*).ti.
- 27 ((online or internet or web* or digital*) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 28 (phone* or telephone* or smartphone* or cellphone* or smartwatch*).ti.
- 29 ((phone* or telephone* or smartphone* or cellphone* or smartwatch*) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 30 (mobile health or mhealth or m-health or ehealth or e-health or emental or e-mental).ti.
- 31 ((mobile health or mhealth or m-health or ehealth or e-health or emental or e-mental) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 32 (mobile* adj3 (based or application* or intervention* or device* or technolog*)).ti,ab.
- 33 text messaging/
- 34 (SMS or text messag*).ti,ab.
- 35 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34
- 36 12 and 35
- 37 7 or 36
- 38 limit 37 to yr="2015 -Current"

Number of results from the first search in Embase:

5145

Number of results from the updated search in Embase:

1174

Total number of results from Embase:

6319

MEDLINE

Ovid MEDLINE(R) ALL 1946 to 2023 August 18

Ovid MEDLINE(R) ALL <1946 to 2023 August 18>

- 1 exp Digital Divide/
- 2 (digital adj3 (divide* or equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or diversity or balance* or unbalance* or imbalance*)).ti,ab.
- 3 1 or 2
- 4 exp Women/or exp Gender Identity/or exp Transgender Persons/or exp Pregnancy/
- 5 (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*).ti,ab.
- 6 4 or 5
- 7 3 and 6
- 8 Gender Equity/or Women's Rights/

- 9 ((female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) adj3 (equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or discrimination* or disaggregat* or diversity or balance* or unbalance* or imbalance* or right*)).ti,ab.
- 10 ((female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) adj3 (use* or usage* or utili#ation or access*)).ti,ab.
- 11 unequal world.ti,ab.
- 12 8 or 9 or 10 or 11
- 13 exp Digital Technology/or exp Information Technology/
- 14 (digital adj3 (technolog* or electronic* or environment* or age or inclusion or era or world)).ti,ab.
- 15 ((information or communication) adj3 technolog*).ti,ab.
- 16 (knowledge society or digiti#ation).ti,ab.
- 17 exp Telemedicine/
- 18 ((tele* or digital or mobile or smart) adj3 health).ti,ab.
- 19 (telehealth or telemedicine or telemonitoring).ti,ab.
- 20 Mobile Applications/
- 21 exp Internet/
- 22 exp Cell Phone/
- 23 exp Computers, Handheld/
- 24 Medical Informatics Applications/
- 25 Therapy, Computer-Assisted/
- 26 (app or apps).ti,ab.
- 27 (online or internet or web* or digital*).ti.
- 28 ((online or internet or web* or digital*) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 29 (phone* or telephone* or smartphone* or cellphone* or smartwatch*).ti.
- 30 ((phone* or telephone* or smartphone* or cellphone* or smartwatch*) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 31 (mobile health or mhealth or m-health or ehealth or e-health or emental or e-mental).ti.
- 32 ((mobile health or mhealth or m-health or ehealth or e-health or emental or e-mental) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 33 (mobile* adj3 (based or application* or intervention* or device* or technolog*)).ti,ab.
- 34 Text Messaging/
- 35 (SMS or text messag*).ti,ab.
- 36 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35
- 37 12 and 36
- 38 7 or 37
- 39 limit 38 to yr="2015 -Current"

Number of results from the first search in MEDLINE:

3601

Number of results from the updated search in MEDLINE:

658

Total number of results from MEDLINE:

4259

American Psychological Association (APA) PsycInfo®

APA PsycInfo 1806 to August 2023

APA PsycInfo <1806 to August 2023>

- 1 exp Digital Divide/
- 2 (digital adj3 (divide* or equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or diversity or balance* or unbalance* or imbalance*)).ti,ab.
- 3 1 or 2
- 4 exp Human Females/or exp gender identity/or exp Pregnancy/
- 5 (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*).ti,ab.
- 6 4 or 5
- 7 3 and 6
- 8 exp Gender Equality/or exp Gender Gap/
- 9 ((female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) adj3 (equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or discrimination* or disaggregat* or diversity or balance* or unbalance* or imbalance* or right*)).ti,ab.
- 10 ((female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) adj3 (use* or usage* or utili#ation or access*)).ti,ab.
- 11 unequal [world.ti](#),ab.
- 12 8 or 9 or 10 or 11
- 13 exp "information and communication technology"/
- 14 (digital adj3 (technolog* or electronic* or environment* or age or inclusion or era or world)).ti,ab.
- 15 ((information or communication) adj3 technolog*).ti,ab.
- 16 (knowledge society or digiti#ation).ti,ab.
- 17 exp computer assisted therapy/
- 18 ((tele* or digital or mobile or smart) adj3 health).ti,ab.
- 19 (telehealth or telemedicine or telemonitoring).ti,ab.
- 20 exp mobile applications/
- 21 internet/
- 22 exp mobile devices/
- 23 (app or apps).ti,ab.
- 24 (online or internet or web* or digital*).ti.
- 25 ((online or internet or web* or digital*) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 26 (phone* or telephone* or smartphone* or cellphone* or smartwatch*).ti.
- 27 ((phone* or telephone* or smartphone* or cellphone* or smartwatch*) adj3 (based or application* or intervention* or program* or therap*)).ab.
- 28 (mobile health or mhealth or m-health or ehealth or e-health or emental or e-mental).ti.
- 29 ((mobile health or mhealth or m-health or ehealth or e-health or emental or e-mental) adj3 (based or application* or intervention* or program* or therap*)).ab.

- 30 (mobile* adj3 (based or application* or intervention* or device* or technolog*)).ti,ab.
 31 exp text messaging/
 32 (SMS or text messag*).ti,ab.
 33 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or
 29 or 30 or 31 or 32
 34 12 and 33
 35 7 or 34
 36 limit 35 to yr="2015 -Current"

Number of results from the first search in APA PsycInfo: 2793
Number of results from the updated search in APA PsycInfo: 418
Total number of results from APA PsycInfo: 3211

Cumulative Index to Nursing and Allied Health Literature (CINAHL)

Print search history

Friday, 2023 August 18 8:43:20 AM

Query Limiters/Expanders

- S37 S36 Limiters – Published Date: 20150101-20231231
 S36 S7 OR S35 Expanders – Apply equivalent subjects
 S35 S12 AND S34 Expanders – Apply equivalent subjects
 S34 S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR
 S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 Expanders
 – Apply equivalent subjects
 S33 TI SMS or "text messag*") OR AB SMS or "text messag*") Expanders – Apply
 equivalent subjects
 S32 TI mobile* N3 (based or application* or intervention* or device* or technolog*) OR
 AB mobile* N3 (based or application* or intervention* or device* or technolog*)
 Expanders – Apply equivalent subjects
 S31 AB ("mobile health" or mhealth or m-health or ehealth or e-health or emental or e-mental)
 N3 (based or application* or intervention* or program* or therap*) Expanders – Apply
 equivalent subjects
 S30 TI "mobile health" or mhealth or m-health or ehealth or e-health or emental or e-mental
 Expanders – Apply equivalent subjects
 S29 AB (phone* or telephone* or smartphone* or cellphone* or smartwatch*) N3 (based
 or application* or intervention* or program* or therap*) Expanders – Apply equivalent
 subjects
 S28 TI phone* or telephone* or smartphone* or cellphone* or smartwatch* Expanders – Apply
 equivalent subjects
 S27 AB (online or internet or web* or digital*) N3 (based or application* or intervention* or
 program* or therap*) Expanders – Apply equivalent subjects

- S26 TI online or internet or web* or digital* Expanders – Apply equivalent subjects
- S25 TI app or apps) OR AB app or apps) Expanders – Apply equivalent subjects
- S24 (MH “Computers, Hand-Held+”) Expanders – Apply equivalent subjects
- S23 (MH “Therapy, Computer Assisted+”) Expanders – Apply equivalent subjects
- S22 (MH “Cellular Phone+”) Expanders – Apply equivalent subjects
- S21 (MH “Internet”) Expanders – Apply equivalent subjects
- S20 (MH “Mobile Applications”) Expanders – Apply equivalent subjects
- S19 TI telehealth or telemedicine or telemonitoring) OR AB telehealth or telemedicine or telemonitoring) Expanders – Apply equivalent subjects
- S18 TI (tele* or digital or mobile or smart) N3 health) OR AB (tele* or digital or mobile or smart) N3 health) Expanders – Apply equivalent subjects
- S17 (MH “Telehealth+”) Expanders – Apply equivalent subjects
- S16 TI “knowledge society” or digiti#ation) OR AB “knowledge society” or digiti#ation) Expanders – Apply equivalent subjects
- S15 TI (information or communication) N3 technolog*) OR AB (information or communication) N3 technolog*) Expanders – Apply equivalent subjects
- S14 TI digital N3 (technolog* or electronic* or environment* or age or inclusion or era or world)) OR AB digital N3 (technolog* or electronic* or environment* or age or inclusion or era or world)) Expanders – Apply equivalent subjects
- S13 (MH “Information Technology+”) Expanders – Apply equivalent subjects
- S12 S8 OR S9 OR S10 OR S11 Expanders – Apply equivalent subjects
- S11 TI “unequal world” OR AB “unequal world” Expanders – Apply equivalent subjects
- S10 TI (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) N3 (use* or usage* or utili#ation or access*)) OR AB (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) N3 (use* or usage* or utili#ation or access*)) Expanders – Apply equivalent subjects
- S9 TI (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) N3 (equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or discrimination* or disaggregat* or diversity or balance* or unbalance* or imbalance* or right*) OR AB (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) N3 (equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or discrimination* or disaggregat* or diversity or balance* or unbalance* or imbalance* or right*)) Expanders – Apply equivalent subjects
- S8 (MH “Gender Equality”) OR (MH “Women’s Rights”) Expanders – Apply equivalent subjects
- S7 S3 AND S6 Expanders – Apply equivalent subjects
- S6 S4 OR S5 Expanders – Apply equivalent subjects
- S5 TI (female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) OR AB female* or wom#n* or gender or sex or transgender or transwom#n* or transsexual* or pregnan* or matern*) Expanders – Apply equivalent subjects
- S4 (MH “Female”) OR (MH “Gender Identity+”) OR (MH “Sexual and Gender Minorities+”) OR (MH “Pregnancy+”) Expanders – Apply equivalent subjects
- S3 S1 OR S2 Expanders – Apply equivalent subjects

- S2 TI digital N3 (divide* or equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or diversity or balance* or unbalance* or imbalance*) OR AB digital N3 (divide* or equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or diversity or balance* or unbalance* or imbalance*) Expanders – Apply equivalent subjects
- S1 (MH “Digital Divide”) Expanders – Apply equivalent subjects

Number of results from the first search in CINAHL: 2716
Number of results from the updated search in CINAHL: 320
Total number of results from CINAHL: 3036

Cochrane Library

Date Run: 18-08-2023 11:02

- ID Search
- #1 MeSH descriptor: (Digital Divide) explode all trees
 (digital NEAR/3 (divide* or equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or diversity or balance* or unbalance* or imbalance*)):ti,ab,kw
- #3 #1 or #2
- #4 MeSH descriptor: (Women) explode all trees
- #5 MeSH descriptor: (Gender Identity) explode all trees
- #6 MeSH descriptor: (Transgender Persons) explode all trees
- #7 MeSH descriptor: (Pregnancy) explode all trees
- #8 (female* or wom*n* or gender or sex or transgender or transwom*n* or transsexual* or pregnan* or matern*):ti,ab,kw
- #9 #4 or #5 or #6 or #7 or #8
- #10 #3 and #9
- #11 MeSH descriptor: (Gender Equity) explode all trees
- #12 MeSH descriptor: (Women's Rights) explode all trees
- #13 ((female* or wom*n* or gender or sex or transgender or transwom*n* or transsexual* or pregnan* or matern*) NEAR/3 (equit* or equalit* or inequit* or inequalit* or gap or bias* or difference* or discrimination* or disaggregat* or diversity or balance* or unbalance* or imbalance* or right*)):ti,ab,kw
- #14 ((female* or wom*n* or gender or sex or transgender or transwom*n* or transsexual* or pregnan* or matern*) NEAR/3 (use* or usage* or utili?ation or access*)):ti,ab,kw
- #15 (unequal world):ti,ab,kw
- #16 #11 or #12 or #13 or #14 or #15
- #17 MeSH descriptor: (Digital Technology) explode all trees
- #18 MeSH descriptor: (Information Technology) explode all trees

- #19 (digital NEAR/3 (technolog* or electronic* or environment* or age or inclusion or era or world)):ti,ab,kw
- #20 ((information or communication) NEAR/3 technolog*):ti,ab,kw
- #21 ("knowledge society" or digiti?ation):ti,ab,kw
- #22 MeSH descriptor: (Telemedicine) explode all trees
- #23 ((tele* or digital or mobile or smart) NEAR/3 health):ti,ab,kw
- #24 (telehealth or telemedicine or telemonitoring):ti,ab,kw
- #25 MeSH descriptor: (Mobile Applications) explode all trees
- #26 MeSH descriptor: (Internet) explode all trees
- #27 MeSH descriptor: (Cell Phone) explode all trees
- #28 MeSH descriptor: (Computers, Handheld) explode all trees
- #29 MeSH descriptor: (Medical Informatics Applications) explode all trees
- #30 MeSH descriptor: (Therapy, Computer-Assisted) explode all trees
- #31 (app or apps):ti,ab,kw
- #32 (online or internet or web* or digital*):ti
- #33 ((online or internet or web* or digital*) NEAR/3 (based or application* or intervention* or program* or therap*)):ab
- #34 (phone* or telephone* or smartphone* or cellphone* or smartwatch*):ti
- #35 ((phone* or telephone* or smartphone* or cellphone* or smartwatch*) NEAR/3 (based or application* or intervention* or program* or therap*)):ab
- #36 ("mobile health" or mhealth or m-health or ehealth or e-health or emental or e-mental):ti
- #37 (("mobile health" or mhealth or m-health or ehealth or e-health or emental or e-mental) NEAR/3 (based or application* or intervention* or program* or therap*)):ab
- #38 (mobile* NEAR/3 (based or application* or intervention* or device* or technolog*)):ti,ab,kw
- #39 MeSH descriptor: (Text Messaging) explode all trees
- #40 (SMS or text messag*):ti,ab,kw
- #41 #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40
- #42 #16 and #41
- #43 #10 or #42
- #44 #43 with Publication Year from 2015 to 2022, in Trials

Number of results from the first search in CENTRAL:	1149
Number of results from the updated search in CENTRAL:	204
Total number of results from CENTRAL:	1353

Google Scholar

By: Lasse Østengaard, University Library of Southern Denmark, University of Southern Denmark, Odense, Denmark, and Department of Clinical Research, University of Southern Denmark, Odense, Denmark.

Date: 8 August 2023

Description of the Google Scholar search: four searches were developed for Google Scholar based on the research questions for this review. The searches were conducted through Publish or Perish 8, and the first 300 results from each search were saved and screened for eligibility.

Search 1

Keywords: (Women) AND (telehealth OR telemedicine OR telemonitoring OR “mobile health” OR mhealth OR m-health OR ehealth OR e-health OR emental OR e-mental)

Time limitation: from 2015

Search 2

Keywords: (digital divide health) AND (female OR women OR gender OR sex OR transgender OR transwomen OR transsexual OR pregnant OR pregnancy OR maternal)

Time limitation: from 2015

Search 3

Keywords: (digital health access) AND (female OR women OR gender OR sex OR transgender OR transwomen OR transsexual OR pregnant OR pregnancy OR maternal)

Time limitation: from 2015

Search 4

Keywords: (digital health empowerment) AND (female OR women OR gender OR sex OR transgender OR transwomen OR transsexual OR pregnant OR pregnancy OR maternal)

Time limitation: from 2015



Annex 3.Excluded studies

Excluded studies are listed in Table A3.1 by year and first author.

Table A3.1. Excluded studies²

2015		
Study	Citation	Reason for exclusion
Billings et al., 2015	Billings DW, Leaf SL, Spencer J, Crenshaw T, Brockington S, Dalal RS. A randomized trial to evaluate the efficacy of a web-based HIV behavioral intervention for high-risk African American women. <i>AIDS Behav.</i> 2015;19(7):1263–74. https://doi.org/10.1007/s10461-015-0999-9 . PMID: 25616838; PMCID: PMC4506203.	Study has mixed population or population not related to the review question
Chilukuri et al., 2015	Chilukuri N, West M, Henderson JL, Lawson S, Ehsanipoor R, Costigan K et al. Information and communication technology use among low-income pregnant and postpartum women by race and ethnicity: a cross-sectional study. <i>J Med Internet Res.</i> 2015;17(7):e163. https://doi.org/10.2196/jmir.3916 . PMID: 26142162; PMCID: PMC4526977.	Study has mixed population or population not related to the review question
Clouse et al., 2015	Clouse K, Schwartz SR, Van Rie A, Bassett J, Vermund SH, Pettifor AE. High mobile phone ownership, but low Internet and email usage among pregnant, HIV-infected women attending antenatal care in Johannesburg. <i>J Telemed Telecare.</i> 2015;21(2):104–7. https://doi.org/10.1177/1357633X14566569 . PMID: 25586808; PMCID: PMC4860457.	Study has mixed population or population not related to the review question
Döring & Prinzellner, 2015	Döring N, Prinzellner Y. Girls' peer sex counseling in online discussion forums (abstract). <i>J Sex Med.</i> 2015;12(suppl. 6):322. https://doi.org/10.1111/jsm.12956 .	Study with unclear or ineligible study designs

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Kim et al., 2015	Kim HK, Niederdeppe J, Guillory J, Graham M, Olson C, Gay G. Determinants of pregnant women's online self-regulatory activities for appropriate gestational weight gain. <i>Health Commun.</i> 2015;30(9):922–32. https://doi.org/10.1080/10410236.2014.905900 . PMID: 25205417; PMCID: PMC4535434.	Study has mixed population or population not related to the review question
Lukasse, 2015	Lukasse M. A mobile smartphone application to promote a healthy diet and physical activity among pregnant women with GDM – RCT. Oslo: Oslo Metropolitan University; 2015 (https://clinicaltrials.gov/study/NCT02588729).	Study did not target gender equity and women's empowerment
Pollick et al., 2015	Pollick SA, Beatty JR, Sokol RJ, Strickler RC, Chang G, Svikis DS et al. Acceptability of a computerized brief intervention for alcohol among abstinent but at-risk pregnant women. <i>Subst Abus.</i> 2015;36(1):13–20. https://doi.org/10.1080/08897077.2013.857631 . PMID: 2426677; PMCID: PMC4031312.	Study has mixed population or population not related to the review question
Shalowitz et al., 2015	Shalowitz DI, Smith AG, Bell MC, Gibb RK. Teleoncology for gynecologic cancers. <i>Gynecol Oncol.</i> 2015;139(1):172–7. https://doi.org/10.1016/j.ygyno.2015.06.042 . PMID: 26151077.	Study did not target gender equity and women's empowerment
2016		
Study	Citation	Reason for exclusion
Abdel-Aleem et al., 2016	Abdel-Aleem H, El-Gibaly OMH, El-Gazzar AFE-S, Al-Attar GST. Mobile clinics for women's and children's health. <i>Cochrane Database Syst Rev.</i> 2016;2016(8):CD009677. https://doi.org/10.1002/14651858.CD009677.pub2 . PMID: 27513824; PMCID: PMC9736774.	Study did not target gender equity and women's empowerment
Åhman et al., 2016	Åhman A, Sarkadi A, Lindgren P, Rubertsson C. "It made you think twice" – an interview study of women's perception of a web-based decision aid concerning screening and diagnostic testing for fetal anomalies. <i>BMC Pregnancy Childbirth.</i> 2016;16(1):267. https://doi.org/10.1186/s12884-016-1057-y . PMID: 27619366; PMCID: PMC5020555.	

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Atkinson et al., 2016	Atkinson KM, Westeinde J, Ducharme R, Wilson SE, Deeks SL, Crowcroft N et al. Can mobile technologies improve on-time vaccination? A study piloting maternal use of ImmunizeCA, a pan-Canadian immunization app. <i>Hum Vaccin Immunother</i> . 2016;12(10):2654–61. https://doi.org/10.1080/21645515.2016.1194146 . PMID: 27322109; PMCID: PMC5085001.	Study has mixed population or population not related to the review question
Beatty et al., 2016	Beatty L, Kemp E, Butow P, Girgis A, Schofield P, Turner J et al. What women with metastatic breast cancer want in an internet-intervention for cancer-related distress: a qualitative exploration. <i>Asia Pac J Clin Oncol</i> . 2016;12(suppl. 5):141. https://doi.org/10.1111/ajco.12646 .	Study did not target gender equity and women's empowerment
Berglund Scherwitzl et al., 2016	Berglund Scherwitzl E, Danielsson KG, Sellberg JA, Scherwitzl R. Fertility awareness-based mobile application for contraception. <i>Eur J Contracept Reprod Health Care</i> . 2016;21(3):234–41. https://doi.org/10.3109/13625187.2016.1154143 . PMID: 27003381; PMCID: PMC4898152.	Study has mixed population or population not related to the review question
Bert et al., 2016	Bert F, Passi S, Scaioli G, Gualano MR, Siliquin R. There comes a baby! What should I do? Smartphones' pregnancy-related applications: a web-based overview. <i>Health Informatics J</i> . 2016;22(3):608–17. https://doi.org/10.1177/1460458215574120 . PMID: 25900813.	Study has mixed population or population not related to the review question
Blackstock et al., 2016	Blackstock OJ, Cunningham CO, Haughton LJ, Garner RY, Norwood C, Horvath KJ. Higher ehealth literacy is associated with HIV risk behaviors among HIV-infected women who use the internet. <i>J Assoc Nurses AIDS Care</i> . 2016;27(1):102–8. https://doi.org/10.1016/j.jana.2015.09.001 . PMID: 26456347; PMCID: PMC4695307.	Study has mixed population or population not related to the review question
Farrant & Heazall, 2016	Farrant K, Heazall AEP. Online information for women and their families regarding reduced fetal movements is of variable quality, readability and accountability. <i>Midwifery</i> . 2016;34:72–8. https://doi.org/10.1016/j.midw.2015.12.013 . PMID: 26867705.	Study has mixed population or population not related to the review question

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Harrington, 2016	Harrington EK. Evaluation of an mHealth SMS dialogue strategy to meet women's and couples' postpartum contraceptive needs in Kenya (Mobile WACH XY). Seattle (WA): University of Washington; 2016 (https://clinicaltrials.gov/show/NCT02781714).	Study did not target gender equity and women's empowerment
Mullaney et al., 2016	Mullaney L, O'Higgins AC, Cawley S, Kennedy R, McCartney D, Turner MJ. Technology based dietary assessment tool in early pregnancy (abstract). Obes Facts. 2016;9(suppl. 1):137. https://doi.org/10.1159/000446744 . PMID: 27238363; PMCID: PMC5672850.	Study with unclear or ineligible study designs
Osma et al., 2016	Osma J, Barrera AZ, Ramphos E. Are pregnant and postpartum women interested in health-related apps? Implications for the prevention of perinatal depression. Cyberpsychol Behav Soc Netw. 2016;19(6):412–5. https://doi.org/10.1089/cyber.2015.0549 . PMID: 27327069.	Study has mixed population or population not related to the review question
Sondaal et al., 2016	Sondaal SFV, Browne JL, Amoakoh-Coleman M, Borgstein A, Miltenburg AS, Verwijs M et al. Assessing the effect of mhealth interventions in improving maternal and neonatal care in low- and middle-income countries: a systematic review. PLoS One. 2017;11(5):e0154664. https://doi.org/10.1371/journal.pone.0154664 . PMID: 27144393; PMCID: PMC4856298.	Study did not target gender equity and women's empowerment

2017

Study	Citation	Reason for exclusion
Ag Ahmed et al., 2017	Ag Ahmed MA, Gagnon M-P, Hamelin-Brabant L, Mbemba GIC, Alami H. A mixed methods systematic review of success factors of mhealth and telehealth for maternal health in Sub-Saharan Africa. Mhealth. 2017;3:22. https://doi.org/10.21037/mhealth.2017.05.04 . PMID: 28736731; PMCID: PMC5505928.	Study did not target gender equity and women's empowerment

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Aiken et al., 2017	Aiken ARA, Digol I, Trussell J, Gomperts R. Self reported outcomes and adverse events after medical abortion through online telemedicine: population based study in the Republic of Ireland and Northern Ireland. <i>BMJ</i> . 2017;357:j2011. https://doi.org/10.1136/bmj.j2011 . PMID: 28512085; PMCID: PMC5431774.	Study has mixed population or population not related to the review question
Bahanshal et al., 2017	Bahanshal S, Coughlin S, Liu B. For you and your baby (4YYB): adapting the centers for disease control and prevention's text4baby program for Saudi Arabia. <i>JMIR Res Protoc</i> . 2017;6(2):e23. https://doi.org/10.2196/resprot.5818 . PMID: 28246065; PMCID: PMC5350457.	Study has mixed population or population not related to the review question
Bush et al., 2017	Bush J, Barlow DE, Echols J, Wilkerson J, Bellevin K. Impact of a mobile health application on user engagement and pregnancy outcomes among Wyoming Medicaid members. <i>Telemed J E Health</i> . 2017;23(11):891–8. https://doi.org/10.1089/tmj.2016.0242 PMID: 28481167; PMCID: PMC5684663.	Study has mixed population or population not related to the review question
Gray et al., 2017	Gray J, Beatty JR, Svikis DS, Puder KS, Resnicow K, Konkel J et al. Electronic brief intervention and text messaging for marijuana use during pregnancy: initial acceptability of patients and providers. <i>JMIR Mhealth Uhealth</i> . 2017;5(11):e172. https://doi.org/10.2196/mhealth.7927 . PMID: 29117931; PMCID: PMC5700401.	Study has mixed population or population not related to the review question
Guendelman et al., 2017	Guendelman S, Broderick A, Mlo H, Gemmill A, Lindema D. Listening to communities: mixed-method study of the engagement of disadvantaged mothers and pregnant women with digital health technologies. <i>J Med Internet Res</i> . 2017;19(7):e240. https://doi.org/10.2196/jmir.7736 . PMID: 28679489; PMCID: PMC5517821.	Study has mixed population or population not related to the review question

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Kirby, 2017	Kirby KL. Empowerment processes in the lives of Tanzanian women: intersection of family, education, and digital technology (Doctoral dissertation, Kent State University). Columbus (OH): OhioLINK Electronic Theses and Dissertations Center; 2017 (http://rave.ohiolink.edu/etdc/view?acc_num=kent1478777628317389).	Study did not target gender equity and women's empowerment
Lehavot et al., 2017	Lehavot K, Litz B, Millard SP, Hamilton AB, Sadler A, Simpson T. Study adaptation, design, and methods of a web-based PTSD intervention for women veterans. <i>Contemp Clin Trials</i> . 2017;53:68–79. https://doi.org/10.1016/j.cct.2016.12.002 . PMID: 27940187.	Study has mixed population or population not related to the review question
Masika, 2017	Masika R. Mobile phones and entrepreneurial identity negotiation by urban female street traders in Uganda. <i>Gend Work Organ</i> . 2017;24(6):610–27. https://doi.org/10.1111/gwao.12184 .	Study did not target gender equity and women's empowerment
Sommer et al., 2017	Sommer J, Daus M, Smith M, Luna D. Mobile application for pregnant women: what do mothers say? <i>Stud Health Technol Inform</i> . 2017;245:221–4. PMID: 29295086.	Study has mixed population or population not related to the review question
Watson et al., 2017	Watson HA, Carter J, Seed PT, Tribe RM, Shennan AH. The QUIPP app: a safe alternative to a treat-all strategy for threatened preterm labor. <i>Ultrasound Obstet Gynecol</i> . 2017; 50(3):342–6. https://doi.org/10.1002/uog.17499 . PMID: 28436125.	Study has mixed population or population not related to the review question
2018		
Study	Citation	Reason for exclusion
Abejirinde et al., 2018	Abejirinde I-OO, Douwes R, Bardaji A, Abugnaba-Abanga R, Zweekhorst, van Roosmale J et al. Pregnant women's experiences with an integrated diagnostic and decision support device for antenatal care in Ghana. <i>BMC Pregnancy Childbirth</i> . 2018;18:209. https://doi.org/10.1186/s12884-018-1853-7 . PMID: 29871596; PMCID: PMC5989381.	Study has mixed population or population not related to the review question

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Abejirinde et al., 2018	Abejirinde I-OO, Ilozumba O, Marchal B, Zweekhorst M, Dieleman M. Mobile health and the performance of maternal health care workers in low- and middle-income countries: a realist review. 2018;21(3):73–86. https://doi.org/10.1177/2053434518779491 . PMID: 30271609; PMCID: PMC6151957.	Study did not target gender equity and women's empowerment
Ainscough et al., 2018	Ainscough K, Kennelly MA, O'Sullivan EJ, Lindsay KL, Gibney ER, McCarthy M et al. Impact of a smartphone app supporting a lifestyle intervention in overweight and obese pregnancy on maternal health and lifestyle outcomes. Am J Obstet Gynecol. 2018;218(1 suppl.):s598–9. https://doi.org/10.1016/j.ajog.2017.11.551 .	Study with unclear or ineligible study designs
AlJaberi, 2018	AlJaberi H. Developing culturally sensitive mhealth apps for Caribbean immigrant women to use during pregnancy: focus group study. JMIR Hum Factors. 2018;5(4):e29. https://doi.org/10.2196/humanfactors.9787 . PMID: 30305256; PMCID: PMC6231776.	Study has mixed population or population not related to the review question
Basir et al., 2018	Basir M, Ahamed I (Sheikh), Jones C, Dreier S, Barnekow K, Basir M et al. The preemie prep for parents (P3) mobile app – a new approach to educating expectant parents who are at risk for preterm delivery. Pediatrics. 2018;141(1_ meetingabstract):17. https://doi.org/10.1542/peds.141.1MA1.17 .	Study with unclear or ineligible study designs
Bliss et al., 2018	Bliss DZ, Gurvick OV, Patel S, Meyer I, Richter HE. Self-management of anal incontinence and interest in a supportive m-health app among women. Neurourol Urodyn. 2018;37 (suppl. 5):S67–8.	Study with unclear or ineligible study designs
Bonnell et al., 2018	Bonnell S, Griggs A, Avila G, Mack J, Bush RA, Vignato J et al. Community health workers and use of mhealth: improving identification of pregnancy complications and access to care in the Dominican Republic. Health Promot Pract. 2018;19(3):331–40. https://doi.org/10.1177/1524839917708795 . PMID: 28578606.	Study has mixed population or population not related to the review question

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Dodd et al., 2018	Dodd JM, Louise J, Cramp C, Grivell RM, Moran LJ, Deussen AR. Evaluation of a smartphone nutrition and physical activity application to provide lifestyle advice to pregnant women: the SNAPP randomised trial. <i>Matern Child Nutr.</i> 2018;14(1):e12502. https://doi.org/10.1111/mcn.12502 . PMID: 28836373; PMCID: PMC6866107.	Study has mixed population or population not related to the review question
Flinders University, 2018	Online prevention of disordered eating in at-risk young-adult women: a two-country pragmatic randomised controlled trial. In: International Clinical Trials Registry Platform [website]. Geneva: World Health Organization; 2018 (https://trialsearch.who.int/Trial2.aspx?TrialID=ACTRN12618001156279).	Study did not target gender equity and women's empowerment
Garnweidner-Holme et al., 2018	Garnweidner-Holme L, Andersen TH, Sando MW, Noll J, Lukasse M. Health care professionals' attitudes toward, and experiences of using, a culture-sensitive smartphone app for women with gestational diabetes mellitus: qualitative study. <i>JMIR Mhealth Uhealth.</i> 2018;6(5):e123. https://doi.org/10.2196/mhealth.9686 . PMID: 29759959; PMCID: PMC5972202.	Study has mixed population or population not related to the review question
Hughson et al., 2018	Hughson J-AP, Daly JO, Woodward-Kron R, Hajek J, Story D. The rise of pregnancy apps and the implications for culturally and linguistically diverse women: narrative review. <i>JMIR Mhealth Uhealth.</i> 2018;6(11):e189. https://doi.org/10.2196/mhealth.9119 . PMID: 30446483; PMCID: PMC6269626.	Study did not target gender equity and women's empowerment
Hunter Medical Research Unit, 2018	Postnatal support app for first time mothers about maternal self-efficacy. In: International Clinical Trials Registry Platform [website]. Geneva: World Health Organization; 2018 (https://trialsearch.who.int/Trial2.aspx?TrialID=ACTRN12618001580268).	Study did not target gender equity and women's empowerment

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Liverpool Hospital Department of Obstetrics and Gynaecology, 2018	My exercise my offspring (MEMO) study: a randomised controlled trial to evaluate the effectiveness of a physical activity program for pregnant women. In: International Clinical Trials Registry Platform [website]. Geneva: World Health Organization; 2018 (https://trialsearch.who.int/Trial2.aspx?TrialID=ACTRN12618000159257).	Study did not target gender equity and women's empowerment
Mo et al., 2018	Mo Y, Gong W, Wang J, Sheng X, Xu DR. The association between the use of antenatal care smartphone apps in pregnant women and antenatal depression: cross-sectional study. JMIR Mhealth Uhealth. 2018;6(11):e11508. https://doi.org/10.2196/11508 . PMID: 30497996; PMCID: PMC6293246.	Study has mixed population or population not related to the review question
Omole et al., 2018	Omole O, Ijadunola MY, Olotu E, Omotoso O, Bello B, Awoniran O et al. The effect of mobile phone short message service on maternal health in south-west Nigeria. Int J Health Plann Manage. 2018;33(1):155–70. https://doi.org/10.1002/hpm.2404 . PMID: 28332259.	Study has mixed population or population not related to the review question
Reyes et al., 2018	Reyes J, Washio Y, Stringer M, Teitelman AM. Usability and acceptability of Everhealthier Women, a mobile application to enhance informed health choices. J Obstet Gynecol Neonatal Nurs. 2018;47(6):853–61. https://doi.org/10.1016/j.jogn.2018.04.139 . PMID: 30415655.	Study has mixed population or population not related to the review question
Sharp, 2018	Sharp T. Improving equity in rural maternity care with technology. Women Birth. 2018;31(suppl. 1):s44. https://doi.org/10.1016/j.wombi.2018.08.131 .	Study with unclear or ineligible study designs
Steege et al., 2018	Steege R, Waldman L, Datiko DG, Kea AZ, Taegtmeier M, Theobald S. "The phone is my boss and my helper" – a gender analysis of an mhealth intervention with health extension workers in Southern Ethiopia. J Public Health (Oxf). 2018;40(suppl. 2):ii16–31. https://doi.org/10.1093/pubmed/fdy199 . PMID: 30551130; PMCID: PMC6294039.	Study did not target gender equity and women's empowerment

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Tobe et al., 2018	Tobe RG, Haque SE, Ikegami K, Mori R. Mobile-health tool to improve maternal and neonatal health care in Bangladesh: a cluster randomized controlled trial. <i>BMC Pregnancy Childbirth</i> . 2018;18(1):102. https://doi.org/10.1186/s12884-018-1714-4 . PMID: 29661178; PMCID: PMC5902947.	Study has mixed population or population not related to the review question
Yamin et al., 2018	Yamin F, Kaewkungwal J, Singhasivanon P, Lawpoolsri S. Women's perceptions of using mobile phones for maternal and child health support in Afghanistan: cross-sectional survey. <i>JMIR Mhealth Uhealth</i> . 2018;6(4):e76. https://doi.org/10.2196/mhealth.9504 . PMID: 29636317; PMCID: PMC5915672.	Study has mixed population or population not related to the review question
2019		
Study	Citation	Reason for exclusion
Abuidhail et al., 2019	Abuidhail J, Mrayan L, Jaradat D. Evaluating effects of prenatal web-based breastfeeding education for pregnant mothers in their third trimester of pregnancy: Prospective randomized control trial. <i>Midwifery</i> . 2019;69:143–9. https://doi.org/10.1016/j.midw.2018.11.015 . PMID: 30513445.	Study has mixed population or population not related to the review question
Acquavita et al., 2019	Acquavita SP, Krummel DA, Talks A, Cobb A, McClure E. Assessing the digital divide among low-income perinatal women: opportunities for provision of health information and counselling. <i>Telemed J E Health</i> . 2019;25(1):48–54. https://doi.org/10.1089/tmj.2017.0292 . PMID: 29708865; PMCID: PMC6913108.	Study has mixed population or population not related to the review question
Afutendem et al., 2019	Afutendem NB, Baleba AN, Tapondjou AL, Nkfusai CN, Viyoff VZ, Nsai FS et al. Assessing implementation modalities of mhealth intervention on pregnant women in Dschang health district, west region of Cameroon. <i>Pan Afr Med J</i> . 2019;33:305. https://doi.org/10.11604/pamj.2019.33.305.17603 . PMID: 31723372; PMCID: PMC6842446.	Study has mixed population or population not related to the review question

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Alqudah et al., 2019	Alqudah A, McMullan P, Todd A, O'Doherty C, McVey A, McConnell M et al. Service evaluation of diabetes management during pregnancy in a regional maternity hospital: potential scope for increased self-management and remote patient monitoring through mhealth solutions. <i>BMC Health Serv Res.</i> 2019;19(1):662. https://doi.org/10.1186/s12913-019-4471-9 . PMID: 31514743; PMCID: PMC6743173.	Study has mixed population or population not related to the review question
Amoakoh et al., 2019	Amoakoh HB, Klipstein-Grobusch K, Grobbee DE, Amoakoh-Coleman M, Oduro-Mensah E, Sarpong C et al. Using mobile health to support clinical decision-making to improve maternal and neonatal health outcomes in Ghana: insights of frontline health worker information needs. <i>JMIR Mhealth Uhealth.</i> 2019;7(5):e12879. https://doi.org/10.2196/12879 . PMID: 31127719; PMCID: PMC6555115.	Study has mixed population or population not related to the review question
Angus et al., 2019	Angus K, Dunne L, Gledhill S, Miles S. Breastfeeding apps: tool of empowerment, or just a hi-tech diary? <i>Women Birth.</i> 2019;32(suppl. 1):s4–5. https://doi.org/10.1016/j.wombi.2019.07.165 .	Study with unclear or ineligible study designs
Asklund & Samuelsson, 2019	Asklund I, Samulesson E. The use of an app with a PFMT programme among pregnant and postnatal women for preventive use and treatment of urinary incontinence. <i>Neurourol Urodyn.</i> 2019;38(suppl. 3):s452–3.	Study with unclear or ineligible study designs
Bajracharya & Dangol, 2019	Bajracharya B, Dangol M. Impact of mass media exposure in getting HIV testing among urban women in Nepal. <i>Sex Transm Infect.</i> 2019;95(suppl. 1):a95. https://doi.org/10.1136/sextrans-2019-sti.243 .	Study with unclear or ineligible study designs
Chan & Chen, 2019	Chan KL, Chen M. Effects of social media and mobile health apps on pregnancy care: meta-analysis. <i>JMIR Mhealth Uhealth.</i> 2019;7(1):e11836. https://doi.org/10.2196/11836 . PMID: 30698533; PMCID: PMC6372934.	Study did not target gender equity and women's empowerment

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Chyzzy & Dennis, 2019	Chyzzy B, Dennis C-L. Mobile phone-based peer support in the prevention of postpartum depression among adolescent mothers: a pilot randomized controlled trial. <i>J Adolesc Health</i> . 2019;64(2 suppl.):s8–9. https://doi.org/10.1016/j.jadohealth.2018.10.030 .	Study with unclear or ineligible study designs
Deave et al., 2019	Deave T, Ginja S, Goodenough T, Bailey E, Piwek L, Coad J et al. The Bumps and Babies Longitudinal Study (BaBBLeS): a multi-site cohort study of first-time mothers to evaluate the effectiveness of the Baby Buddy app. <i>Mhealth</i> . 2019;5:42. https://doi.org/10.21037/mhealth.2019.08.05 . PMID: 31620469; PMCID: PMC6789295.	Study has mixed population or population not related to the review question
Dehlin et al., 2019	Dehlin JM, Stillwagon R, Pickett J, Keene L, Schneider JA. #PrEP4Love: an evaluation of a sex-positive HIV prevention campaign. <i>JMIR Public Health Surveill</i> . 2019;5(2):e12822. https://doi.org/10.2196/12822 . PMID: 31210141; PMCID: PMC6601419.	Study did not target gender equity and women's empowerment
Gasteiger et al., 2019	Gasteiger N, Anderson A, Day K. Rethinking engagement: exploring women's technology use during the perinatal period through a Kaupapa Māori consistent approach. <i>J N Z Coll Midwives</i> . 2019;55:20–26. https://doi.org/10.12784/nzcomjnl55.2019.3.20-26 .	Study has mixed population or population not related to the review question
Gill et al., 2019	Gill R, Ogilvie G, Norman WV, Fitzsimmons B, Maher C, Renner R. Feasibility and acceptability of a mobile technology intervention to support postabortion care in British Columbia: phase I. <i>J Med Internet Res</i> . 2019;21(5):e13387. https://doi.org/https://doi.org/10.2196/13387 . PMID: 31144668 PMCID: PMC6658228	Study has mixed population or population not related to the review question
Harrington et al., 2019	Harrington EK, Drake A, Matemo D, Ronen K, Osoti AO, John-Stewart G et al. An mhealth SMS intervention on postpartum contraceptive use among women and couples in Kenya: a randomized controlled trial. <i>Am J Public Health</i> . 2019;109(6):934–41. https://doi.org/10.2105/AJPH.2019.305051 . PMID: 31067089 PMCID: PMC6507993.	Study has mixed population or population not related to the review question

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Holt et al., 2019	Holt J, Cusatis R, Asan O, Williams JS, Flynn KE, Crotty BH. Can patients humanize their electronic health record to facilitate stronger connections with care teams? <i>J Gen Intern Med.</i> 2019;34(2 suppl.):S710–1. https://dx.doi.org/10.1007/11606.1525-1497 .	Study did not target gender equity and women's empowerment
Kozak et al., 2019	Kozak M, Usoz M, Fujimoto D, Von Eyben R, Kidd EA. Phase I pilot study evaluating email and telephone reminders for vaginal dilator use following vaginal brachytherapy. <i>Int J Radiat Oncol Biol Phys.</i> 2019;105(1 suppl.):e344. https://doi.org/10.1016/j.ijrobp.2019.06.1758 .	Study with unclear or ineligible study designs
Laitinen et al., 2019	Laitinen K, Merikivi A, Muhli E, Koivuniemi E. Use of mobile app for health promotion in normal weight and overweight or obese pregnant women. <i>Obes Facts.</i> 2019;12(suppl. 1):130. https://dx.doi.org/10.1159/000489691 .	Study did not target gender equity and women's empowerment
Lally et al., 2019	Lally RM, Bellavia G, Gallo S, Kupzyk K, Helgeson V, Brooks C et al. Feasibility and acceptance of the CaringGuidance web-based, distress self-management, psychoeducational program initiated within 12 weeks of breast cancer diagnosis. <i>Psychooncology.</i> 2019;28(4):888–95. https://doi.org/10.1002/pon.5038 . PMID: 30803084.	Study has mixed population or population not related to the review question
Masoi & Kibusi, 2019	Masoi TJ, Kibusi SM. Improving pregnant women's knowledge on danger signs and birth preparedness practices using an interactive mobile messaging alert system in Dodoma region, Tanzania: a controlled quasi experimental study. <i>Reprod Health.</i> 2019;16(1):177. https://doi.org/10.1186/s12978-019-0838-y . PMID: 31831076; PMCID: PMC6909441.	Study has mixed population or population not related to the review question
Monash University, 2019	The study for mother-infant sleep (the SMILE project): reducing postpartum insomnia in first-time mothers. In: International Clinical Trials Registry Platform [website]. Geneva: World Health Organization; 2018 (https://trialsearch.who.int/Trial2.aspx?TrialID=ACTRN12619001166167).	Study did not target gender equity and women's empowerment

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Propper, 2019	Propper, de Moura O. The relationship between social media use and maternal self-efficacy in primiparous mothers (DPsych dissertation). Boston (MA): William James College. 2019 (https://www.proquest.com/openview/852f06d707244dc9cce8c92fe843c281/1?pq-origsite=gscholar&cbl=18750&diss=y).	Study did not target gender equity and women's empowerment
Sreehari & Pillai, 2019	Sreehari S, Pillai S. Innovative technology for breast cancer death prevention in the developing world: a global cancer care perspective. J Pathol. 2019;248(suppl. 1):S5. https://dx.doi.org/10.1002/path.5276 .	Study with unclear or ineligible study designs
Sundstrom et al., 2019	Sundstrom B, DeMaria AL, Ferrara M, Meier S, Billings D. "The closer, the better": the role of telehealth in increasing contraceptive access among women in rural South Carolina. Matern Child Health J. 2019;23(9):1196–1205. https://doi.org/10.1007/s10995-019-02750-3 . PMID: 31228142.	Study has mixed population or population not related to the review question
Tang et al., 2019	Tang S, Ghose B, Hoque MK, Hao G, Yaya S. Women using mobile phones for health communication are more likely to use prenatal and postnatal services in Bangladesh: cross-sectional study. JMIR Mhealth Uhealth. 2019;7(2):e10645. https://doi.org/10.2196/10645 . PMID: 30816850; PMCID: PMC6416540.	Study has mixed population or population not related to the review question
van Beukering et al., 2019	van Beukering M, Velu A, van den Berg L, Kok M, Mol BW, Frings-Dresen M et al. Usability and usefulness of a mobile health app for pregnancy-related work advice: mixed-methods approach. JMIR Mhealth Uhealth. 2019;7(5):e11442. https://doi.org/10.2196/11442 . PMID: 31094353; PMCID: PMC6532337.	Study has mixed population or population not related to the review question
Wang et al., 2019	Wang NA, Deng Z, Wen LM, Ding Y, He G. Understanding the use of smartphone apps for health information among pregnant Chinese women: mixed methods study. JMIR Mhealth Uhealth. 2019;7(6):e12631. https://doi.org/10.2196/12631 . PMID: 31215516; PMCID: PMC6604500.	Study has mixed population or population not related to the review question

Table A3.1 contd

Zaleski et al., 2019	Zaleski K, Enrile A, Weiss EL, Wang X, editors. Women's journey to empowerment in the 21st century: a transnational feminist analysis of women's lives in modern times. Oxford: Oxford Academic; 2020 (https://doi.org/10.1093/oso/9780190927097.001.0001).	Study did not target gender equity and women's empowerment
2020		
Study	Citation	Reason for exclusion
Alam et al., 2020	Alam M, Banwell C, Lokuge K. The effect of women's differential access to messages on their adoption of mobile health services and pregnancy behavior in Bangladesh: retrospective cross-sectional study. JMIR Mhealth Uhealth. 2020;8(7):e17665. https://doi.org/10.2196/17665 . PMID: 32706694; PMCID: PMC7399959.	Study has mixed population or population not related to the review question
Allen & Vottero, 2020	Allen J, Vottero B. Experiences of homeless women in accessing health care in community-based settings: a qualitative systematic review. JBI Evid Synth. 2020;18(9):1970–2010. https://doi.org/10.11124/JBISRIIR-D-19-00214 . PMID: 32813421.	Study did not target gender equity and women's empowerment
Ampt et al., 2020	Ampt FH, Lim MSC, Agius PA, L'Engle K, Manguro G, Gichuki C et al. Effect of a mobile phone intervention for female sex workers on unintended pregnancy in Kenya (WHISPER or SHOUT): a cluster-randomised controlled trial. Lancet Glob Health. 2020;8(12):e1534–45. https://doi.org/10.1016/S2214-109X(20)30389-2 . PMID: 33220217.	Study has mixed population or population not related to the review question
Blackwell et al., 2020	Blackwell TM, Dill LJ, Hoepner LA, Geer LA. Using text messaging to improve access to prenatal health information in urban African American and Afro-Caribbean immigrant pregnant women: mixed methods analysis of Text4baby usage. JMIR Mhealth Uhealth. 2020;8(2):e14737. https://doi.org/10.2196/14737 . PMID: 32053117; PMCID: PMC7055759.	Study has mixed population or population not related to the review question

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Bland et al., 2020	Bland C, Dalrymple KV, White SL, Moore A, Poston L, Flynn AC. Smartphone applications available to pregnant women in the United Kingdom: an assessment of nutritional information. <i>Matern Child Nutr.</i> 2020;16(2):e12918. https://doi.org/10.1111/mcn.12918 . PMID: 31833237; PMCID: PMC7083499.	Study did not target gender equity and women's empowerment
Boileau et al., 2020	Boileau N, Margueritte F, Gauthier T, Boukeffa N, Preux P-M, Labrunie A et al. Mobile phone use during pregnancy: which association with fetal growth? <i>J Gynecol Obstet Hum Reprod.</i> 2020;49(8):101852. https://doi.org/10.1016/j.jogoh.2020.101852 . PMID: 32623065.	Study has mixed population or population not related to the review question
Chen et al., 2020	Chen M, Liu X, Zhang J, Sun G, Gao Y, Shi Y et al. Characteristics of online medical care consultation for pregnant women during the COVID-19 outbreak: cross-sectional study. <i>BMJ Open.</i> 2020;10(11):e043461. https://doi.org/10.1136/bmjopen-2020-043461 . PMID: 33203639; PMCID: PMC7674021.	Study has mixed population or population not related to the review question
Li et al., 2020	Li M, Wu X, Sun G, Peng M. Determinants of maternal behavior of mobile phone use during pregnancy. <i>J Healthc Eng.</i> 2020;2020:9465019. https://doi.org/10.1155/2020/9465019 . PMID: 33149872; PMCID: PMC7603607.	Study has mixed population or population not related to the review question
Meltzer-Brody & Kimmel, 2020	Meltzer-Brody S, Kimmel M. The promise of telepsychiatry to reduce maternal mortality by increasing access to maternal mental health and addiction services. <i>Obstet Gynecol.</i> 2020;136(4):643–4. https://doi.org/10.1097/AOG.0000000000004116 . PMID: 32925618.	Study did not target gender equity and women's empowerment
Mohan et al., 2020	Mohan D, Bashingwa JJH, Tiffin N, Dhar D, Mulder N, George A et al. Does having a mobile phone matter? Linking phone access among women to health in India: an exploratory analysis of the National Family Health Survey. <i>PLoS One.</i> 2020;15(7):e0236078. https://doi.org/10.1371/journal.pone.0236078 . PMID: 32687527 PMCID: PMC7371204.	Study has mixed population or population not related to the review question

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Nyati-Jokomo et al., 2020	Nyati-Jokomo Z, Dabengwa IM, Makacha L, Nyapwere N, Dube YP, Chikoko L et al. RoadMApp: a feasibility study for a smart travel application to improve maternal health delivery in a low resource setting in Zimbabwe. BMC Pregnancy Childbirth. 2020;20(1):501. https://doi.org/10.1186/s12884-020-03200-7 . PMID: 32867716; PMCID: PMC7457488.	Study has mixed population or population not related to the review question
Paterno et al., 2020	Paterno MT, Leckenby D, Pachucki MC. Challenges and lessons learned from digital storytelling with parenting women in recovery. Health Promot Pract. 2020;21(3):340–3. https://doi.org/10.1177/1524839920902751 . PMID: 32046517.	Study did not target gender equity and women's empowerment
Tumuhimbise et al., 2020	Tumuhimbise W, Atukunda EC, Ayebaza S, Katusiime J, Mugenyi G, Pinkwart N et al. Maternal health-related barriers and the potentials of mobile health technologies: qualitative findings from a pilot randomized controlled trial in rural Southwestern Uganda. J Family Med Prim Care. 2020;9(7):3657–62. https://doi.org/10.4103/jfmprc.jfmprc_281_20 . PMID: 33102346; PMCID: PMC7567246.	Study has mixed population or population not related to the review question
University of Auckland, 2020	Kowhai study: the Younger Women's Wellness Program after Breast Cancer – a feasibility study of an e-health enabled lifestyle modification intervention. In: International Clinical Trials Registry Platform [website]. Geneva: World Health Organization; 2020 (https://trialsearch.who.int/Trial2.aspx?TrialID=ACTRN12620000260921).	Study did not target gender equity and women's empowerment
University of Sydney, 2020	Smart mums with smart phones 2 (SMs2): text messaging support for women after gestational diabetes. In: International Clinical Trials Registry Platform [website]. Geneva: World Health Organization; 2020 (https://trialsearch.who.int/Trial2.aspx?TrialID=ACTRN12620000615987).	Study did not target gender equity and women's empowerment

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Wang et al., 2020	Wang J, Rogge AA, Armour M, Smith CA, DAdamo CA, Pischke CR et al. International ResearchKit app for women with menstrual pain: development, access, and engagement. <i>JMIR Mhealth Uhealth</i> . 2020;8(2):e14661. https://doi.org/10.2196/14661 . PMID: 33102346; PMCID: PMC7567246.	Study has mixed population or population not related to the review question
Watterson et al., 2020	Watterson JL, Castaneda D, Caricia Catalani C. Promoting antenatal care attendance through a text messaging intervention in Samoa: quasi-experimental study. <i>JMIR Mhealth Uhealth</i> . 2020;8(6):e15890. https://doi.org/10.2196/15890 . PMID: 32484446; PMCID: PMC7298629.	Study has mixed population or population not related to the review question
Yee et al., 2020	Yee LM, Jackson J, Leziak K, Niznik CM, Saber R, Yeh C et al. SweetMama: usability testing of a novel mobile application for diabetes education and support during pregnancy. <i>Am J Obstet Gynecol</i> . 2020;22(1 suppl.):s474–5. https://doi.org/10.1016/j.ajog.2019.11.764 .	Study with unclear or ineligible study designs
2021		
Study	Citation	Reason for exclusion
Acar & Oskay, 2020	Acar Z, Oskay Ü. Mobile health applications used in perinatal period and nursing roles. <i>J Educ Res Nurs</i> . 2021;18(4):451–6. https://doi.org/10.5152/jern.2021.60343 .	Study has mixed population or population not related to the review question
Ackroyd et al., 2021	Ackroyd S, Kim J, Lee N. Telemedicine utilization in gynecologic cancers: understanding the digital divide. <i>Gynecol Oncol</i> . 2021;162(suppl. 1):s286–7. https://doi.org/10.1016/S0090-8258(21)01195-1 .	Study with unclear or ineligible study designs
Al-Rshoud et al., 2021	Al-Rshoud F, Qudsi A, Naffa FW, Al Omari B, AlFalah AG. The use and efficacy of mobile fertility-tracking applications as a method of contraception: a survey. <i>Curr Obstet Gynecol Rep</i> . 2021;10(2):25–9. https://doi.org/10.1007/s13669-021-00305-4 . PMID: 33815936; PMCID: PMC8009922.	Study has mixed population or population not related to the review question

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Al-Sheyab et al., 2021	Al-Sheyab NA, Al Nsour M, Khader YS, Yousif H, Alyahya MS, Taha H et al. Midwives and women's perspectives on family planning in Jordan: human rights, gender equity, decision-making and power dynamics. <i>Heliyon</i> . 2021;7(8):e07810. https://doi.org/10.1016/j.heliyon.2021.e07810 . PMID: 34458635; PMCID: PMC8379452.	Study without clear definition of digital health intervention or no digital technology being assessed
Alvarez Gutierrez & Ventura, 2021	Alvarez Gutierrez S, Ventura AK. Associations between maternal technology use, perceptions of infant temperament, and indicators of mother-to-infant attachment quality. <i>Early Hum Dev</i> . 2021;154:105305. https://doi.org/10.1016/j.earlhumdev.2021.105305 . PMID: 33508559.	Study has mixed population or population not related to the review question
Barreto et al., 2021	Barreto, IC de Holanda Cunha, Soares Barros NB, Theophilo RL, Viana VF, de Vasconcelos Silveira FR, de Souza O et al. Development and evaluation of the GISSA Mother-Baby chatbot application in promoting child health. <i>Cien Saude Colet</i> . 2021;26(5):1679–90. https://doi.org/10.1590/1413-81232021265.04072021 . PMID: 34076110.	Study has mixed population or population not related to the review question
Beckham et al., 2021	Beckham SW, Stockton M, Galai N, Davis W, Mwambo J, Likindikoki S et al. Family planning use and correlates among female sex workers in a community empowerment HIV prevention intervention in Iringa, Tanzania: a case for tailored programming. <i>BMC Public Health</i> . 2021;21(1):1377. https://doi.org/10.1186/s12889-021-11426-z . PMID: 34247614; PMCID: PMC8274018.	Study without clear definition of digital health intervention or no digital technology being assessed
Bogale et al., 2021	Bogale B, Mørkrid K, Abbas E, Ward IA, Anaya F, Ghanem B et al. The effect of a digital targeted client communication intervention on pregnant women's worries and satisfaction with antenatal care in Palestine – a cluster randomized controlled trial. <i>PLoS One</i> . 2021;16(4):e0249713. https://doi.org/10.1371/journal.pone.0249713 . PMID: 33891597; PMCID: PMC8064599.	Study has mixed population or population not related to the review question

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Chae et al., 2021	Chae S, Lee Y-J, Han H-R. Sources of health information, technology access, and use among non-English-speaking immigrant women: descriptive correlational study. <i>J Med Internet Res.</i> 2021;23(10):e29155. https://doi.org/10.2196/29155 . PMID: 34714249; PMCID: PMC8590186.	Study has mixed population or population not related to the review question
Chappell et al., 2021	Chappell L, Tucker K, You Y-M, Mcmanus R, Bump Study Group. Self-monitoring of blood pressure in women with pregnancy hypertension: the BUMP2 multicentre randomised controlled trial. <i>J Hypertens.</i> 2021;39:e9. https://doi.org/10.1097/01.hjh.0000744452.18575.74 .	Study with unclear or ineligible study designs
Healy, 2021	Healy RL. Zuckerberg, get out of my uterus! An examination of fertility apps, data-sharing and remaking the female body as a digitalized reproductive subject. <i>J Gend Stud.</i> 2021;30(4):406–16. https://doi.org/10.1080/09589236.2020.1845628 .	Study did not target gender equity and women's empowerment
Kyaw Tha Tun et al., 2021	Kyaw Tha Tun E, Nagel J, Bosbach A, Bock S, Kielblock B, Siegmund-Schultze E et al. Telephone-based peer support intervention to reduce depressive symptoms in women with coronary heart disease, a randomized controlled trial in Germany. <i>Women Health.</i> 2021;61(7):619–32. https://doi.org/10.1080/03630242.2021.1953208 . PMID: 34281485.	Study has mixed population or population not related to the review question
Mohan et al., 2021	Mohan D, Scott K, Shah N, Bashingwa JJH, Chakraborty A, Ummer O et al. Can health information through mobile phones close the divide in health behaviours among the marginalised? An equity analysis of Kilkari in Madhya Pradesh, India. <i>BMJ Glob Health.</i> 2021;6(suppl. 5):e005512. https://doi.org/10.1136/bmjgh-2021-005512 . PMID: 34312154; PMCID: PMC8327823.	Study did not target gender equity and women's empowerment

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Musiimenta et al., 2021	Musiimenta A, Tumuhimbise W, Pinkwart N, Katusiime J, Mugenyi G, Atukunda EC et al. A mobile phone-based multimedia intervention to support maternal health is acceptable and feasible among illiterate pregnant women in Uganda: qualitative findings from a pilot randomized controlled trial. Digit Health. 2021;7:2055207620986296. https://doi.org/10.1177/2055207620986296 . PMID: 33717497; PMCID: PMC7917428.	Study has mixed population or population not related to the review question
Nagpal et al., 2021	Nagpal TS, Everest C, Goudreau AD, Manicks M, Adamo KB. To HIIT or not to HIIT? The question pregnant women may be searching for online: a descriptive observational study. Perspect Public Health. 2021;141(2):81–8. https://doi.org/10.1177/1757913920985898 . PMID: 33579178.	Study has mixed population or population not related to the review question
Peng, 2021	Peng AY. A techno-feminist analysis of beauty app development in China's high-tech industry. J Gend Stud. 2021;30(5):596–608. https://doi.org/10.1080/09589236.2021.1929091 .	Study has mixed population or population not related to the review question
Pervin et al., 2021	Pervin J, Sarker BK, Nu UT, Khatun F, Quaiyum Rahman AM, Venkateswaran M et al. Developing targeted client communication messages to pregnant women in Bangladesh: a qualitative study. BMC Public Health. 2021;21(1):759. https://doi.org/10.1186/s12889-021-10811-y . PMID: 33879108; PMCID: PMC8056650.	Study has mixed population or population not related to the review question
Rouhi et al., 2021	Rouhi M, Stirling C, Crisp EP. The “fallacy of normalcy”: a content analysis of women's online post-childbirth health-related support. Women Birth. 2021;34(3):e262–70. https://doi.org/10.1016/j.wombi.2020.04.007 . PMID: 32467050.	Study has mixed population or population not related to the review question
Simas et al., 2021	Simas C, Paterson P, Lees S, Larson HJ. “From my phone, I could rule the world”: critical engagement with maternal vaccine information, vaccine confidence builders and post-Zika outbreak rumours in Brazil. Vaccine. 2021;39(33):4700–4. https://doi.org/10.1016/j.vaccine.2021.06.039 . PMID: 34229891.	Study has mixed population or population not related to the review question

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Vach, 2021	Vach R. Femtech collective seeks a movement toward holistic women's health. For holistic caregivers, technology is filling the large gaps in female health care. In: Chiropractic Economics [website]. Lakewood Ranch (FL); 2021 (https://www.chiroeco.com/femtech-collective/).	Study did not target gender equity and women's empowerment
2022		
Study	Citation	Reason for exclusion
Adedinsewo et al., 2022	Adedinsewo DA, Pollak AW, Phillips SD, Smith TL, Svatikova A, Hayes SN et al. Cardiovascular disease screening in women: leveraging artificial intelligence and digital tools. <i>Circ Res.</i> 2022;130(4):673–90. https://doi.org/10.1161/CIRCRESAHA.121.319876 . PMID: 35175849; PMCID: PMC8889564.	Study has mixed population or population not related to the review question
Akingbade et al., 2022	Akingbade O, Adediran V, Somoye IE, Alade AS, Chow KM. Perception on the usage of mhealth psychoeducational intervention among Nigerian women with breast cancer receiving chemotherapy: a qualitative study (preprint). <i>Research Square</i> ; 2022. https://doi.org/10.21203/rs.3.rs-1237217/v1 .	Study did not target gender equity and women's empowerment
Alam & Khanam, 2022	Alam MZ, Khanam L. Comparison of the young aged and elderly female users' adoption of mhealth services. <i>Health Care Women Int.</i> 2022;43(10–11):1259–83. https://doi.org/10.1080/07399332.2022.2039149 . PMID: 35195507.	Study has mixed population or population not related to the review question
Alfawzan et al., 2022	Alfawzan N, Christen M, Spitale G, Biller-Andorno N. Privacy, data sharing, and data security policies of women's mhealth apps: scoping review and content analysis. <i>JMIR Mhealth Uhealth.</i> 2022;10(5):e33735. https://doi.org/10.2196/33735 . PMID: 35522465; PMCID: PMC9123546.	Study did not target gender equity and women's empowerment

Table A3.1 contd

Algheshairy et al., 2022	Algheshairy RM, Alhomaïd RM, Almujaïdîl MS, Alharbi HF, Alsanei WA. Influence of using food delivery applications on adult Saudi female dietary habits and preferences during COVID-19 lockdown restrictions: attitude survey. <i>Int J Environ Res Public Health</i> . 2022;19(19):12770. https://doi.org/10.3390/ijerph191912770 . PMID: 36232068; PMCID: PMC9566569.	Study has mixed population or population not related to the review question
Barrera et al., 2022	Barrera AZ, Morris SY, Ruiz A. Mothers and babies online course: participant characteristics and behaviors in a web-based prevention of postpartum depression intervention. <i>Front Glob Womens Health</i> . 2022;3:846611. https://doi.org/10.3389/fgwh.2022.846611 . PMID: 35814836; PMCID: PMC9263388.	Study has mixed population or population not related to the review question
Barro et al., 2022	Barro A, Ngangue P, Bationo N, Soubeiga D, Pafadnam Y, Kabore S et al. Barriers and facilitators to the implementation of cell phone interventions to improve the use of family planning services among women in sub-Saharan Africa: a systematic review. <i>Int J Healthc</i> . 2022;8(2):39–51. https://doi.org/10.5430/ijh.v8n2p39 .	Study did not target gender equity and women's empowerment
Bilal et al., 2022	Bilal A, Bathula D, Bränn E, Fransson E, Virk J, Papadopoulos F et al. Mom2B: a study of perinatal health via smartphone application and machine learning methods. <i>Eur Psychiatry</i> . 2022;65(suppl. 1):s574–5. https://doi.org/10.1192/j.eurpsy.2022.1472 .	Study with unclear or ineligible study designs
Bilal Bomani et al., 2022	Bilal Bomani NZ, James KE, Naert MN, Petrozza JC. The impact of telemedicine on minority women's access to gynecologic and fertility care during the COVID-19 pandemic. <i>Fertil Steril</i> . 2022;118(4 suppl.):2369. https://doi.org/10.1016/j.fertnstert.2022.09.196 .	Study with unclear or ineligible study designs
Bojko et al., 2022	Bojko A, Vaughan DA, Sakkas D, Ryley DA, Halperin F. An effective telehealth based multidisciplinary approach to weight loss in women with obesity seeking treatment for infertility. <i>Fertil Steril</i> . 2022;118(4 suppl.):e31–2. https://doi.org/10.1016/j.fertnstert.2022.08.108 .	Study with unclear or ineligible study designs

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Çaka & Çınar, 2022	Çaka SY, Çınar N. Effect of education on the readiness levels of primigravida women to hygienic care practices of the newborn with the use of QR code. <i>Health Care Women Int.</i> 2022;43(12):1380–1400. https://doi.org/10.1080/07399332.2021.1908289 . PMID: 33939938.	Study has mixed population or population not related to the review question
Canfield, 2022	Canfield SM. Improving perinatal mental health: a pilot study of the online mothers and babies course with couples. In: Mospace [website]. Columbia (MO): University of Missouri; 2022 (https://mospace.umsystem.edu/xmlui/handle/10355/85766).	Study has mixed population or population not related to the review question
Carissoli et al., 2022	Carissoli C, Gasparri D, Riva G, Villani D. Mobile well-being in pregnancy: suggestions from a quasi-experimental controlled study. <i>Behav Inf Technol.</i> 2022;41(8):1639–51. https://doi.org/10.1080/0144929X.2021.1894484 .	Study has mixed population or population not related to the review question
Clouse et al., 2022	Clouse K, Noholoza S, Ngcobo N, Madwayi S, Mrubata M, Camlin CS et al. Cohort profile: careConekta: a pilot study of a smartphone application to improve engagement in postpartum HIV care in South Africa. <i>BMJ Open.</i> 2022;12(11):e064946. https://doi.org/10.1136/bmjopen-2022-064946 . PMID: 36414286; PMCID: PMC9685000.	Study has mixed population or population not related to the review question
Crist et al., 2022	Crist KD, Schor S, Levochkina O, Borba C, McMahon J. The role of telehealth in an obstetric-psychiatry clinic: patient perceptions. <i>J Acad Consult Liaison Psychiatry.</i> 2022;63(suppl. 2):S212. https://dx.doi.org/10.1016/j.jaclp.2022.10.206 .	Study with unclear or ineligible study designs
Del Busso et al., 2022	Del Busso L, Brottveit G, Løkkeberg ST, Gluppe G. Women's embodied experiences of using wearable digital self-tracking health technology: a review of the qualitative research literature. <i>Health Care Women Int.</i> 2022;43(12):1355–79. https://doi.org/10.1080/07399332.2021.1884682 . PMID: 33900152.	Study did not target gender equity and women's empowerment

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Eaves et al., 2022	Eaves ER, Doerry E, Lanzetta SA, Kruthoff KM, Negron K et al. Applying user-centered design in the development of a supportive mhealth app for women in substance use recovery. <i>Am J Health Promot.</i> 2023;37(1):56–64. https://doi.org/10.1177/08901171221113834 . PMID: 35815770; PMCID: PMC9771986.	Study has mixed population or population not related to the review question
El Ayadi et al., 2022	El Ayadi AM, Duggal M, Bagga R, Singh P, Kumar V, Ahuja A et al. A mobile education and social support group intervention for improving postpartum health in northern India: development and usability study. <i>JMIR Form Res.</i> 2022;6(6):e34087. https://doi.org/10.2196/34087 . PMID: 35767348; PMCID: PMC9280461.	Study has mixed population or population not related to the review question
Hensel et al., 2022	Hensel DJ, Von Hippel CD, Sandidge R, Lapage CC, Zelin NS, Perkins RH. "OMG, yes!": feasibility, acceptability, and preliminary efficacy of an online intervention for female sexual pleasure. <i>J Sex Res.</i> 2022;59(3):269–82. https://doi.org/10.1080/00224499.2021.1912277 . PMID: 34176390.	Study has mixed population or population not related to the review question
Hoey & Reynolds, 2022	Hoey W, Reynolds K. Midwifery and obstetric emergency telehealth service across Country WA aims to improve patient access, outcomes and staff retention. <i>Women Birth.</i> 2022;35(suppl. 1):30. https://doi.org/10.1016/j.wombi.2022.07.083 .	Study with unclear or ineligible study designs
Kassa & Matlakala, 2022	Kassa A, Matlakala MC. Effectiveness of mhealth application at primary health care to improve maternal and new-born health services in rural Ethiopia: comparative study (preprint). <i>medRxiv.</i> 2022. https://doi.org/10.1101/2022.04.02.22272628 .	Study has mixed population or population not related to the review question
Löjdahl et al., 2022	Löjdahl E, Lindam A, Asklund I. App-based pelvic floor muscle training in pregnant and postnatal women: a prospective cohort study exploring factors associated with prevention and improvement of urinary incontinence. <i>Health Sci Rep.</i> 2022;5(5):e781. https://doi.org/10.1002/hsr2.781 . PMID: 36000079 PMCID: PMC938790.	Study has mixed population or population not related to the review question

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Lu et al., 2022	Lu L, Soyemi E, Leziak K, Niznik CM, Simon MA, Grobman WA et al. Exploring psychosocial burdens of diabetes in pregnancy and the role of technology-based support. 2022;226(1 suppl.):s183–4. https://doi.org/10.1016/j.ajog.2021.11.318 .	Study with unclear or ineligible study designs
McCall et al., 2022	McCall T, Threats M, Pillai M, Lakdawala A, Bolton CS. Recommendations for design of a mobile application to support management of anxiety and depression among Black American women. Front Digit Health. 2022;4:1028408. https://doi.org/10.3389/fdgth.2022.1028408 . PMID: 36620185; PMCID: PMC9816326.	Study has mixed population or population not related to the review question
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Pillarisetti & Pillarisetti, 2022	Pillarisetti S, Pillarisetti RR. Developing the mobile app “ABCs of breast health” – using digital innovation to improve breast health in India. Eur J Surg Oncol. 2022;48(5):e240. https://doi.org/10.1016/j.ejso.2022.03.210 .	Study with unclear or ineligible study designs
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Rivers & Robles-Rodríguez, 2022	Rivers J, Robles-Rodríguez E. Evaluation of a three-part equity intervention for women of color with breast cancer. <i>J Clin Oncol</i> . 2022;40(16 suppl.):TPS1129. https://doi.org/10.1200/JCO.2022.40.16_suppl.TPS1129 .	Study did not target gender equity and women's empowerment
Rodriguez et al., 2022	Rodriguez JA, Charles JP, Bates DW, Lyles CR, Southworth B, Samal L. Digital health equity in primary care: implementing a pilot digital health navigator program. <i>J Gen Inter Med</i> . 2022;37(suppl. 2):S259–60. https://dx.doi.org/10.1007/s11606-022-07653-8 .	Study did not target gender equity and women's empowerment
Sadeghijoola et al., 2022	Sadeghijoola N, Afshari P, Jofreh MG, Haghighizadeh MH, Abedi P. Comparing the effects of face-to-face versus phone counseling based on cognitive-behavioral therapy for vasomotor symptoms in postmenopausal women: a randomized controlled trial. <i>Postep Psychiatr Neurol</i> . 2022;31(3):114–20. https://doi.org/10.5114/ppn.2022.122499 . PMID: 37082224; PMCID: PMC9946366.	Study without clear definition of digital health intervention or no digital technology being assessed
Shija et al., 2022	Shija L, Yokoyana J, Massawe N. Empowering women to access maternal healthcare services through a digital momcare wallet. <i>BMC Proc</i> . 2022;16(suppl 2):2. https://doi.org/10.1186/s12919-022-00231-0 .	Study with unclear or ineligible study designs

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University of New South Wales, Sydney – Rural Clinical Campus, 2022	The impact of an online integrative antenatal education programme on labour and birth outcomes. In: International Clinical Trials Registry Platform [website]. Geneva: World Health Organization; 2018 (https://trialsearch.who.int/Trial2.aspx?TrialID=ACTRN12622000609752).	Study did not target gender equity and women's empowerment
Walker et al., 2022	Walker RE, Quong S, Olivier P, Wu L, Xie J, Boyle J. Understanding preconception women's needs and preferences for digital health resources: qualitative study. <i>JMIR Form Res.</i> 2022;6(8):e39280. https://doi.org/10.2196/39280 . PMID: 35930344; PMCID: PMC9391970.	Study has mixed population or population not related to the review question
Wang et al., 2022	Wang X, Shi J, Lee KM. The digital divide and seeking health information on smartphones in Asia: survey study of ten countries. <i>J Med Internet Res.</i> 2022;24(1):e24086. https://doi.org/10.2196/24086 . PMID: 35023845; PMCID: PMC8796039.	Study has mixed population or population not related to the review question
Watts et al., 2022	Watts A, Hassan Z, Coromac-Medrano J, Thangwaritorn P, Losinski G. Self-diagnosis by Google: a qualitative study of sources of menopause information. <i>Menopause.</i> 2022;29(12):1488–9. https://dx.doi.org/10.1097/GME.0000000000002112 .	Study with unclear or ineligible study designs

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Yee et al., 2022	Yee LM, Lu L, Soyemi E, Niznik CM, Jackson J, Leziak K et al. Evaluating the user experience of a novel smartphone application for low-income pregnant individuals with diabetes. <i>Am J Obstet Gynecol.</i> 2022;226(1 suppl.):s440–1. https://doi.org/10.1016/j.ajog.2021.11.732 .	Study with unclear or ineligible study designs
Yee et al., 2022	Yee LM, Niznik CM, Jackson J, Leziak K, Saber R, Yeh C et al. A feasibility randomized trial of a smartphone application for low-income pregnant individuals with diabetes. <i>Am J Obstet Gynecol.</i> 2022;226(1 suppl.):s300–1. https://doi.org/10.1016/j.ajog.2021.11.507 .	Study with unclear or ineligible study designs
Yi et al., 2022	Yi S, Burke C, Reilly A, Straube S, Graterol J, Peabody C. Designing and developing a digital equity dashboard for the emergency department. <i>Ann Emerg Med.</i> 2022;80(4 suppl.):S31–2. https://doi.org/10.1002/emp2.12997 . PMID: 37397184; PMCID: PMC10313910.	Study has mixed population or population not related to the review question
Zolkefle et al., 2022	Zolkefle NZN, Nawi ABM, Roslan I'AB, Ridzun NDB, Ibrahim Z, Cob SAC. Women's equality through e-hailing mobile branding application services. <i>J Pharm Negat Results.</i> 2022;13(9):5663–73.	Study has mixed population or population not related to the review question
2023		
Study	Citation	Reason for exclusion
Althouse et al., 2023	Althouse AD, Abebe KZ, Paasche-Orlow MK, Lalama CM, Ferry D, Lance M et al. Design, rationale, and baseline characteristics of a randomized controlled trial evaluating a mobile relational agent to enhance atrial fibrillation self-care. <i>Contemp Clin Trials.</i> 2023;124:107015. https://doi.org/10.1016/j.cct.2022.107015 . PMID: 36402276; PMCID: PMC9839543.	Study did not target gender equity and women's empowerment
Atukunda & Mindra, 2023	Atukunda EC, Mindra A. Mobile health intervention (Support-moms) in antenatal care to improve maternal health in Uganda. In: <i>ClinicalTrials.gov</i> [website]. Bethesda (MD): National Library of Medicine; 2023 (https://clinicaltrials.gov/study/NCT05940831).	Study did not target gender equity and women's empowerment

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Arena et al., 2023	Arena A, Esposti ED, Orsini B, Moro E, Del Forno S, Cocchi L et al. Endometriosis in the time of internet: how web navigation affects women with endometriosis. <i>Ann Med.</i> 2023;55(1):2215537. https://doi.org/10.1080/07853890.2023.2215537 . PMID: 37254517; PMCID: PMC10234130.	Study has mixed population or population not related to the review question
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Axelrod et al., 2023	Axelrod M, Ezra HL, Galler E, Nir O, Barkai G, Ofir K et al. Putting the “fetus back in maternal-fetal telemedicine: a prospective pilot study. <i>Am J Obstet Gynecol.</i> 2023;228(1 suppl.):s544. https://doi.org/10.1016/j.ajog.2022.11.927 .	Study with unclear or ineligible study designs
Bauer et al., 2023	Bauer NH, Hertle D, Schumacher L. Digitale Hebammenbetreuung in der COVID-19-Pandemie in Deutschland – Akzeptanz bei Müttern [Digital midwifery care during the COVID-19 pandemic in Germany – acceptance of mothers]. <i>Bundesgesundheitsblatt Gesundheitsforschung Gesundheitsschutz.</i> 2023;66(3):292–301. https://doi.org/10.1007/s00103-023-03666-8 . PMID: 36763257; PMCID: PMC9912210 (in German).	Study has mixed population or population not related to the review question
Carrandi et al., 2023	Carrandi A, Hu Y, Karger S, Eddy KE, Vogel JP, Harrison CL et al. Systematic review on the cost and cost-effectiveness of mhealth interventions supporting women during pregnancy. <i>Women Birth.</i> 2023;36(1):3–10. https://doi.org/10.1016/j.wombi.2022.03.007 . PMID: 35339412.	Study did not target gender equity and women's empowerment

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Chaléat & Baud, 2023	Chaléat S, Baud D. Gynécologie – obstétrique. FemTech: les nouvelles technologies au service de la santé de la femme [Obstetrics and gynaecology – FemTech: how digital technologies are serving women's health care]. Rev Med Suisse. 2023;19(809–10):28–34. https://doi.org/10.53738/REVMED.2023.19.809-10.28 . PMID: 36660833 (in French).	Study did not target gender equity and women's empowerment
Cooper et al., 2023	Cooper L, Rosen JG, Zhang L, Pelaez D, Olatunde PF, Owczarzak J et al. Exploring the impact of the COVID-19 pandemic on healthcare and substance use service access among women who inject drugs: a qualitative study. Harm Reduct J. 2023;20(1):71. https://doi.org/10.1186/s12954-023-00793-y . PMID: 37296423; PMCID: PMC10251329.	Study has mixed population or population not related to the review question
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Doucette et al., 2023	Doucette JA, Rousseau J, Vondracek H. Addressing health equity through a telehealth maternal-newborn home visit for nurse practitioner students. J Am Assoc Nurse Pract. 2023;35(1):63–70. https://doi.org/10.1097/JXX.0000000000000810 . PMID: 36476637.	Study has mixed population or population not related to the review question

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Huang et al., 2023	Huang A, Rowen T, Chao M, Gibson C, Schembri M, Subak L. A pilot randomized trial of a telehealth pelvic yoga program for women with chronic pelvic pain. <i>J Sex Med.</i> 2023; 20(suppl. 2):qdad061.112. https://doi.org/10.1093/jsxmed/qdad061.112 .	Study with unclear or ineligible study designs
Kodama & Takaki, 2023	Kodama T, Takaki S. Association between smartphone applications for pregnant women and psychological distress in Japan. <i>J Psychosoc Nurs Ment Health Serv.</i> 2023;61(1):32–8. https://doi.org/10.3928/02793695-20220705-04 . PMID: 35858202.	Study has mixed population or population not related to the review question
Koivuniemi et al., 2023	Koivuniemi E, Raats MM, Ollila H, Löyttyniemi E, Laitinen K. Characterising the use, users and effects of a health app supporting lifestyle changes in pregnant women. <i>Br J Nutr.</i> 2023;130(3):433–45. https://doi.org/10.1017/S0007114522003439 . PMID: 36263460.	Study has mixed population or population not related to the review question
Lavoie et al., 2023	Lavoie M-CC, Okui L, Blanco N, Stoebeu K, Magidson JF, Gokatweng G et al. Feasibility and acceptability of peer-delivered interventions using mhealth for PrEP services among adolescent girls and young women in DREAMS program in Botswana. <i>Glob Health Action.</i> 2023;16(1):2231256. https://doi.org/10.1080/16549716.2023.2231256 . PMID: 37462118; PMCID: PMC10355684.	Study has mixed population or population not related to the review question
Maas et al., 2023	Maas VYF, de Feijter M, Passier A, Kant A. The development of a mobile app to inform pregnant women and their healthcare providers about the safety of over-the-counter medicines during pregnancy and breastfeeding. <i>Neurotoxicol Teratol.</i> 2023;98:107214. https://doi.org/10.1016/j.ntt.2023.107214 .	Study with unclear or ineligible study designs

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Michel et al., 2023	Michel A, Minton L, Sullivan C, Fontenot C, Lee M-J. Integrating mobile and portable technologies with in-person and telehealth care to increase perinatal healthcare equity for the unstably housed. <i>J Perinat Neonatal Nurs.</i> 2023;37(1):27-35. https://doi.org/10.1097/JPN.0000000000000705 . PMID: 36707745.	Study did not target gender equity and women's empowerment
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Okonofua et al., 2023	Okonofua F, Ntoimo L, Johnson E, Sombie I, Ojuolape S, Igboin B et al. Texting for life: a mobile phone application to connect pregnant women with emergency transport and obstetric care in rural Nigeria. <i>BMC Pregnancy Childbirth.</i> 2023;23(1):139. https://doi.org/10.1186/s12884-023-05424-9 . PMID: 36882709; PMCID: PMC9993601.	Study has mixed population or population not related to the review question
Robinson-Whelen et al., 2023	Robinson-Whelen S, Hughes RB, Alhusen JL, Beers L, Minard CG, Davidson D. Health information seeking in the digital age: a national survey of women with disabilities. <i>Disabil Rehabil.</i> 2023;45(17):2751-60. https://doi.org/10.1080/09638288.2022.2105960 . PMID: 35916449; PMCID: PMC9892349.	Study has mixed population or population not related to the review question
Shao, 2023	Shao L. The techno-beauty myth: the self-representations of young Chinese women on social media. <i>J Gend Stud.</i> 2023;32(8):964-76. https://doi.org/10.1080/09589236.2023.2207001 .	Study with unclear or ineligible study designs

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Shimpuku et al., 2023	Shimpuku Y, Mwilike B, Mwakawanga D, Ito K, Hirose N, Kubota K. Development and pilot test of a smartphone app for midwifery care in Tanzania: a comparative cross-sectional study. PLoS One. 2023;18(3):e0283808. https://doi.org/10.1371/journal.pone.0283808 . PMID: 37000830; PMCID: PMC10065243.	Study has mixed population or population not related to the review question
Wei et al., 2023	Wei LS, Bennett T-AM, Friedman S, Chuang M, Mehta-Lee S. Impact of telehealth on improving access to pregnancy care in an underserved population. Am J Obstet Gynecol. 2023;228(1):s524. https://doi.org/10.1016/j.ajog.2022.11.896 . PMCID: PMC9822633.	Study with unclear or ineligible study designs
Western Kentucky University, 2023	Mobile application-based exercise intervention for pregnant women. In: ClinicalTrials.gov [website]. Bethesda (MD): National Library of Medicine; 2023 (https://clinicaltrials.gov/study/NCT05849649?id=NCT05849649&rank=1).	Study did not target gender equity and women's empowerment
Yee et al., 2023	Yee LM, Leziak K, Jackson J, Niznik C, Saber R, Yeh C et al. Sweetmama: usability assessment of a novel mobile application among low-income pregnant people to assist with diabetes management and support. Diabetes Spectr. 2023;36(2):171–81. https://doi.org/10.2337/ds22-0014 . PMID: 37193207; PMCID: PMC10182966.	Study with unclear or ineligible study designs





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